# Generalized Additive Models and beyond with P-splines

## Chapter 4

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# 4. Applications

In this section we will present some examples using the smoothing techniques presented previously applied to some real examples.

## Air quality data

We analyze the `data(airquality)` (see `?airquality`) consisting of air quality measurements in New York, from Maty to September 1973. The data frame contains with 154 observations on 6 variables.

- `[,1] Ozone` numeric Ozone (ppb)
- `[,2] Solar.R` numeric Solar R (lang)
- `[,3] Wind` numeric Wind (mph)
- `[,4] Temp` numeric Temperature (degrees F)
- `[,5] Month` numeric Month (1–12)
- `[,6] Day` numeric Day of month (1–31)

For more details `?airquality`

A simple scatterplot shows that some of the variables have a non-linear relationship.

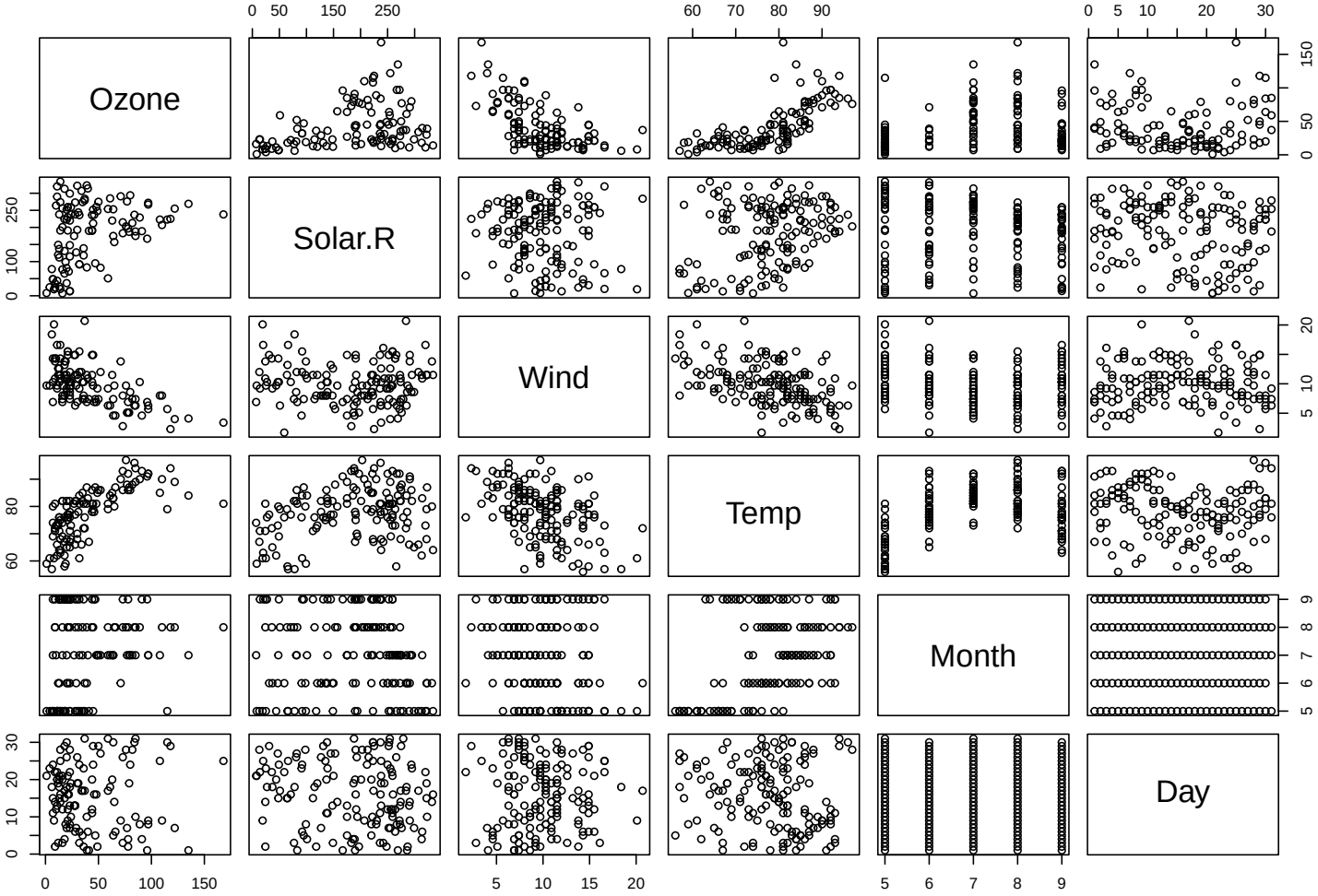



Figure 1: Scatterplot matrix with ‘pairs‘

Let us consider a linear model for the response variable `Ozone`

```r
airq.lm <- lm(Ozone ~ Temp + Wind + Solar.R, data=airquality)
summary(airq.lm)
```

```
##
## Call:
## lm(formula = Ozone ~ Temp + Wind + Solar.R, data = airquality)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -40.485 -14.219  -3.551  10.097  95.619
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -64.34208   23.05472  -2.791  0.00623 **
## Temp          1.65209    0.25353   6.516 2.42e-09 ***
## Wind         -3.33359    0.65441  -5.094 1.52e-06 ***
## Solar.R       0.05982    0.02319   2.580  0.01124 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.18 on 107 degrees of freedom
##   (42 observations deleted due to missingness)
## Multiple R-squared:  0.6059, Adjusted R-squared:  0.5948
## F-statistic: 54.83 on 3 and 107 DF,  p-value: < 2.2e-16
```

Let us plot the results

```
par(mfrow=c(1,3))
termplot(airq.lm,se=TRUE)
```

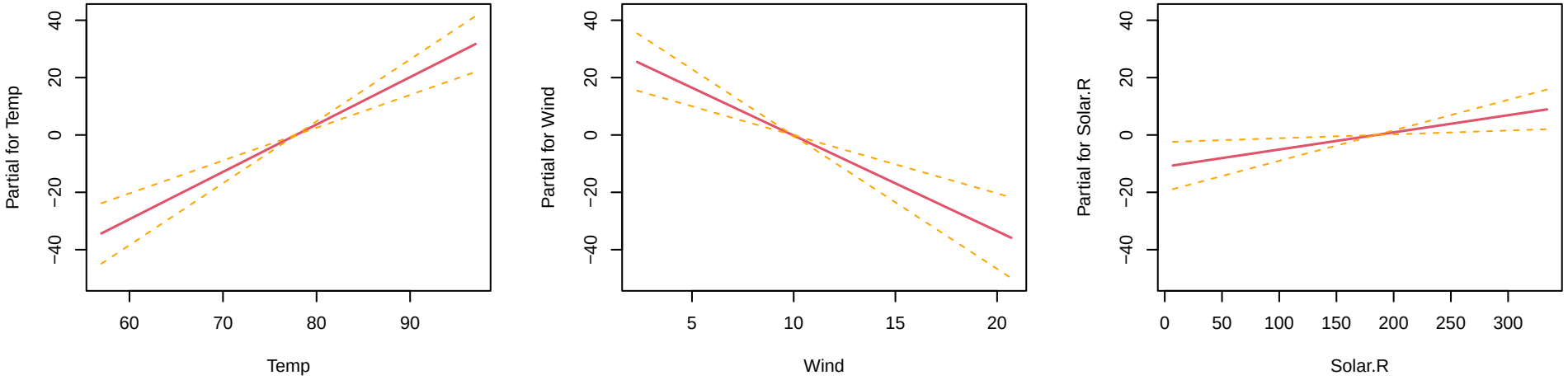



Figure 2: Plots of `lm` fit

We can also plot the residuals of the model in order to check the model hypothesis

```
par(mfrow=c(1,2))
plot(airq.lm,which=1:2)
```

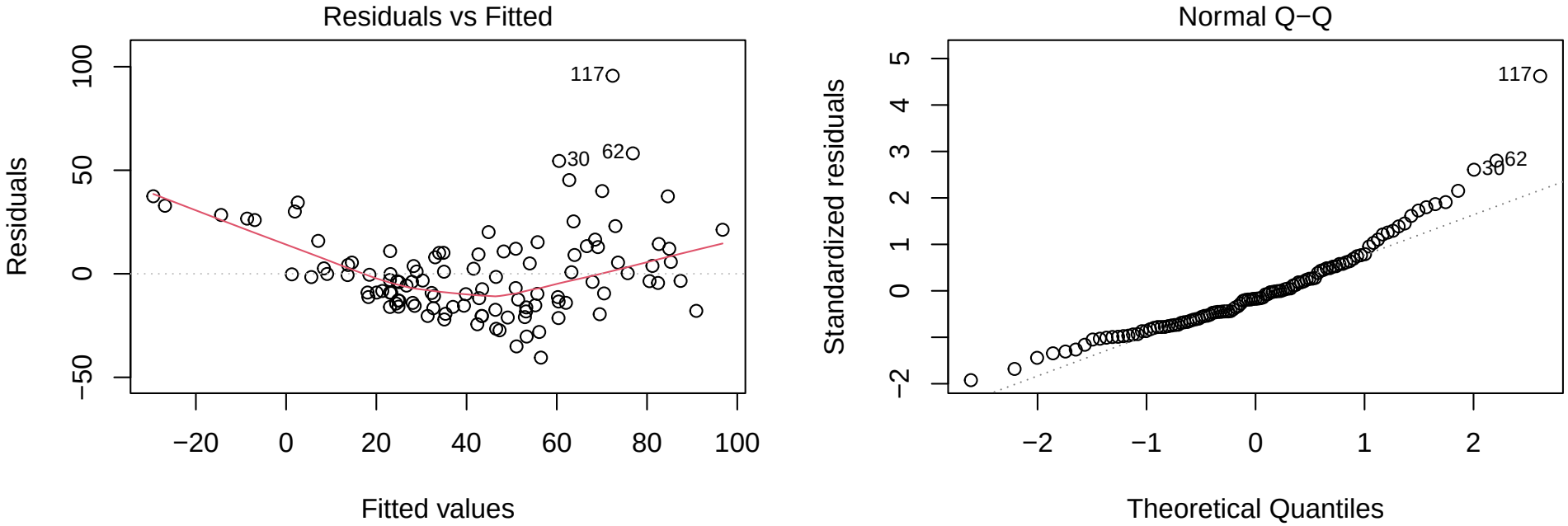



Figure 3: Plots of `lm` residuals

The lack of normality in the residuals is due to the asymmetry of the response variable `Ozone`, we can apply a `log` transform to achieve asymmetry, i.e.:

```
par(mfrow=c(1,2))
hist(airquality$Ozone, main="Ozone")
hist(log(airquality$Ozone), main ="log(Ozone)")
```

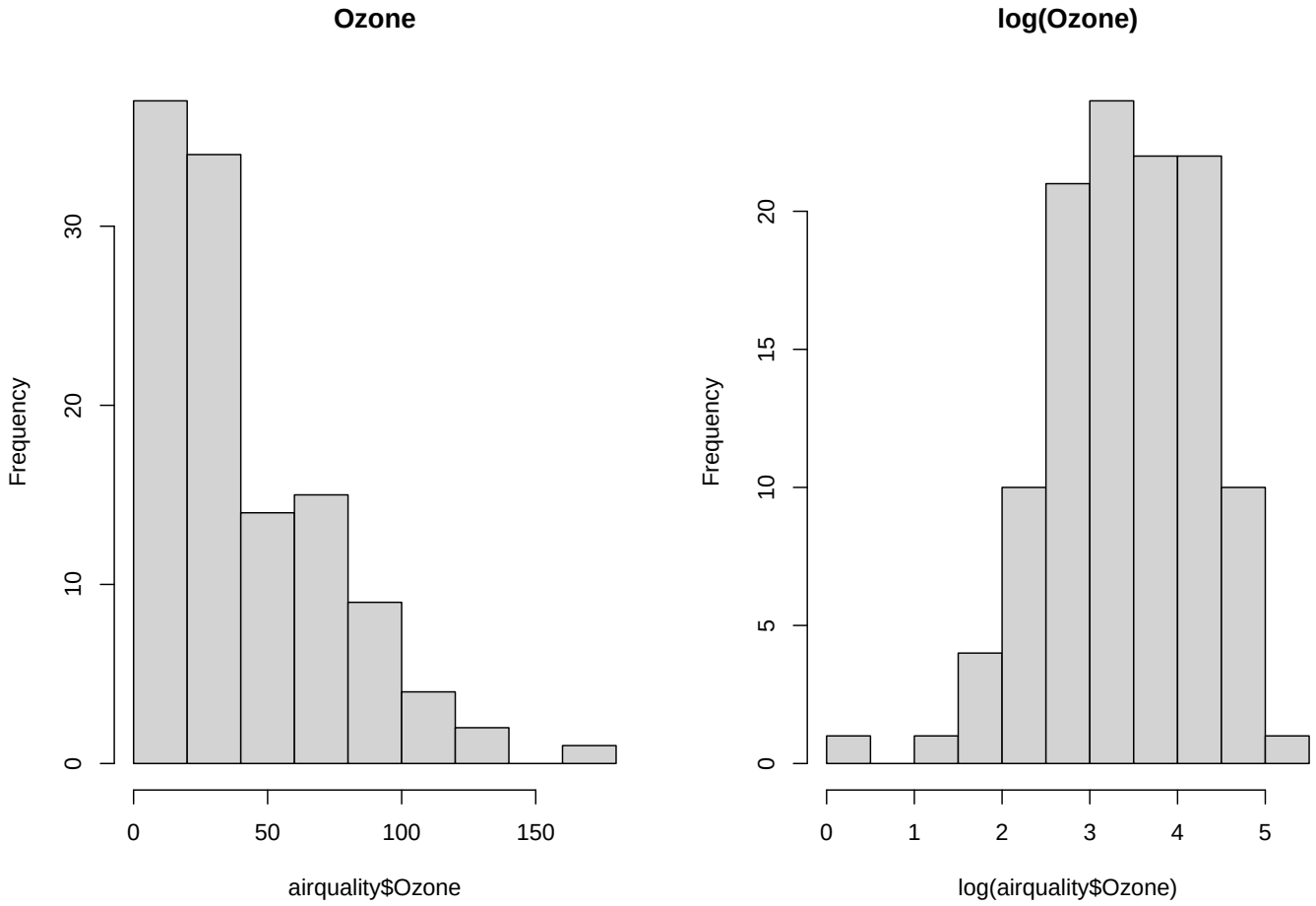
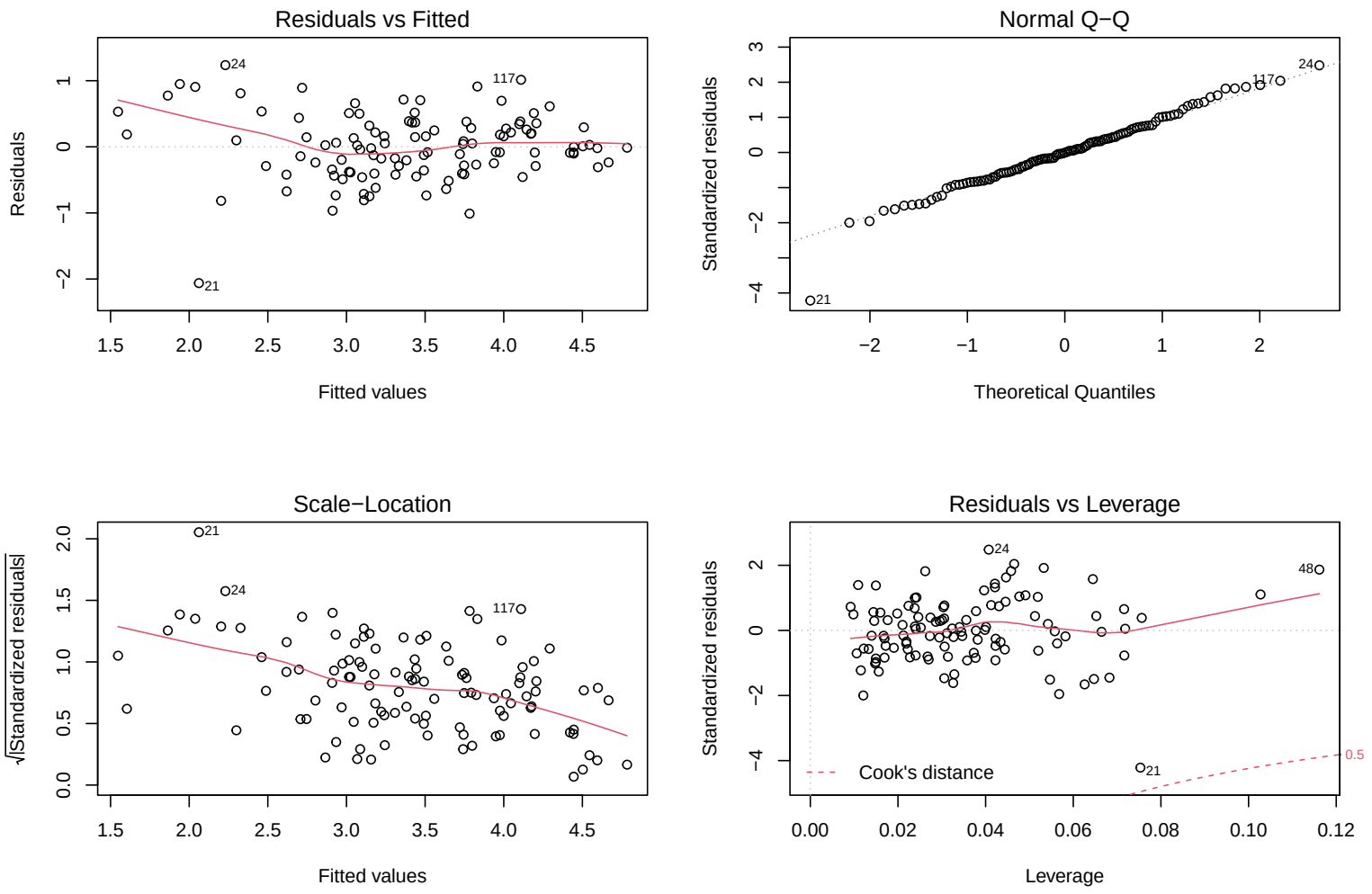



Figure 4: histograms of ‘Ozone‘ and ‘log(Ozone)‘

We can now fit a linear model of the $\log(Ozone)$, does the model look more adecuate?

```
lairq.lm <- lm(log(Ozone)~ Temp + Wind + Solar.R, data=airquality)
summary(lairq.lm)
```

Fitting a linear model we can conclude that there might be some heterokedasticity not accounted by the model. However, the most possible cause is that the relationship between the response variable and the covariates are far from linear.


Residuals vs Fitted
Residuals
Fitted values
24
117
21
Normal Q−Q
Standardized residuals
Theoretical Quantiles
24
117
21
Scale−Location
√|Standardized residuals|
Fitted values
21
24
117
Residuals vs Leverage
Standardized residuals
Leverage
24
48
21
Cook's distance
0.5

# Fitting a GAM

Let us fit a GAM model, firstly with a single variable, i.e,:

```
library(mgcv)
airq.gam1 <- gam(log(Ozone) ~ s(Wind,bs="ps",m=2,k=10),
                 method="REML", select=TRUE,data=airquality)
summary(airq.gam1)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## log(Ozone) ~ s(Wind, bs = "ps", m = 2, k = 10)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.4185     0.0655   52.19   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##           edf Ref.df     F p-value
## s(Wind) 2.565      9 6.453  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.336   Deviance explained =   35%
## -REML = 128.69  Scale est. = 0.49769   n = 116
```

The results show that the smooth effect `s(Wind)` is significative (with an approximate $p$-value close to zero and `edf` $2.56$). The next Figure shows the smooth wind effect. The increment of the wind speed decreases the $\log$ Ozone levels (dramatically until 10mph). Note that including the intercept, the smooth effects are centered at zero.

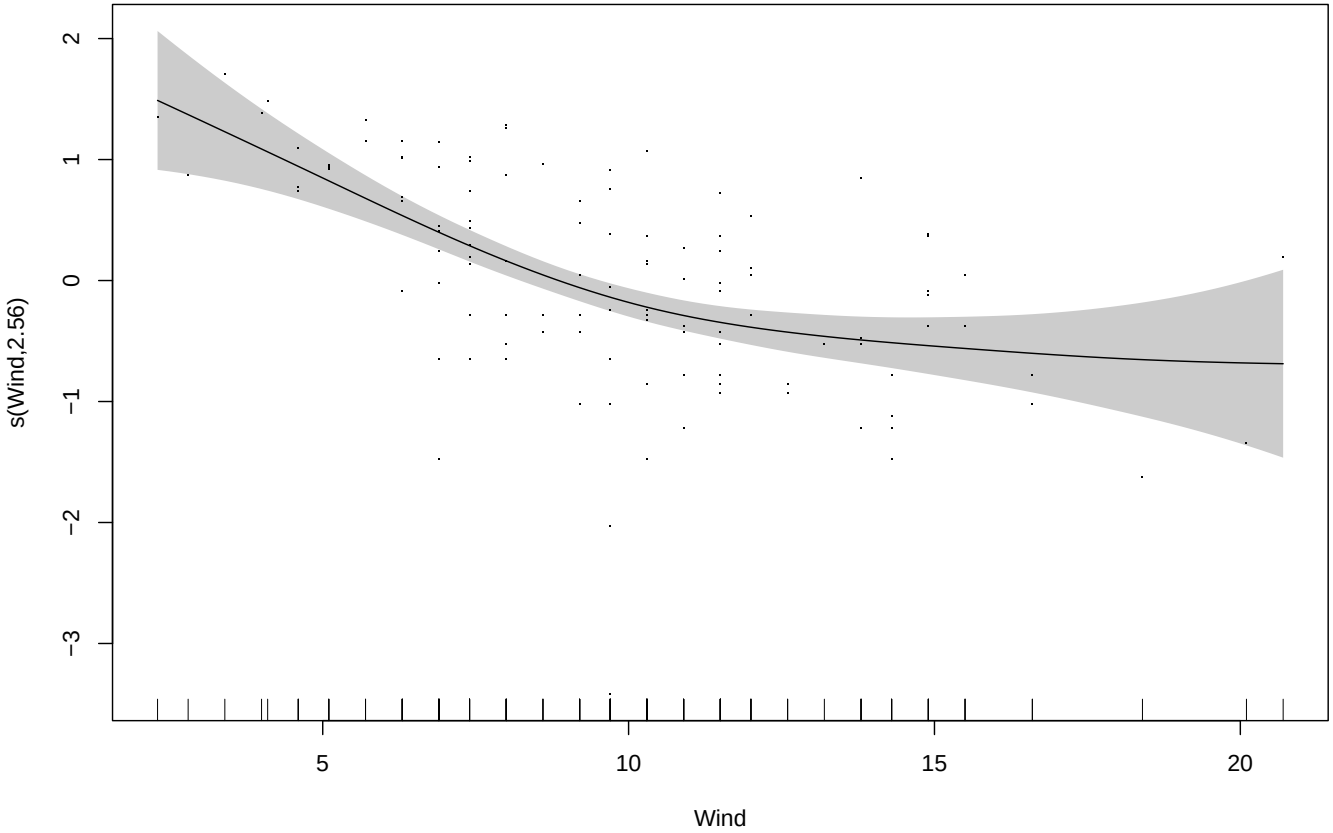



Figure 5: Estimated smooth effect of ‘Wind‘ on ‘log(Ozone)‘. residuals

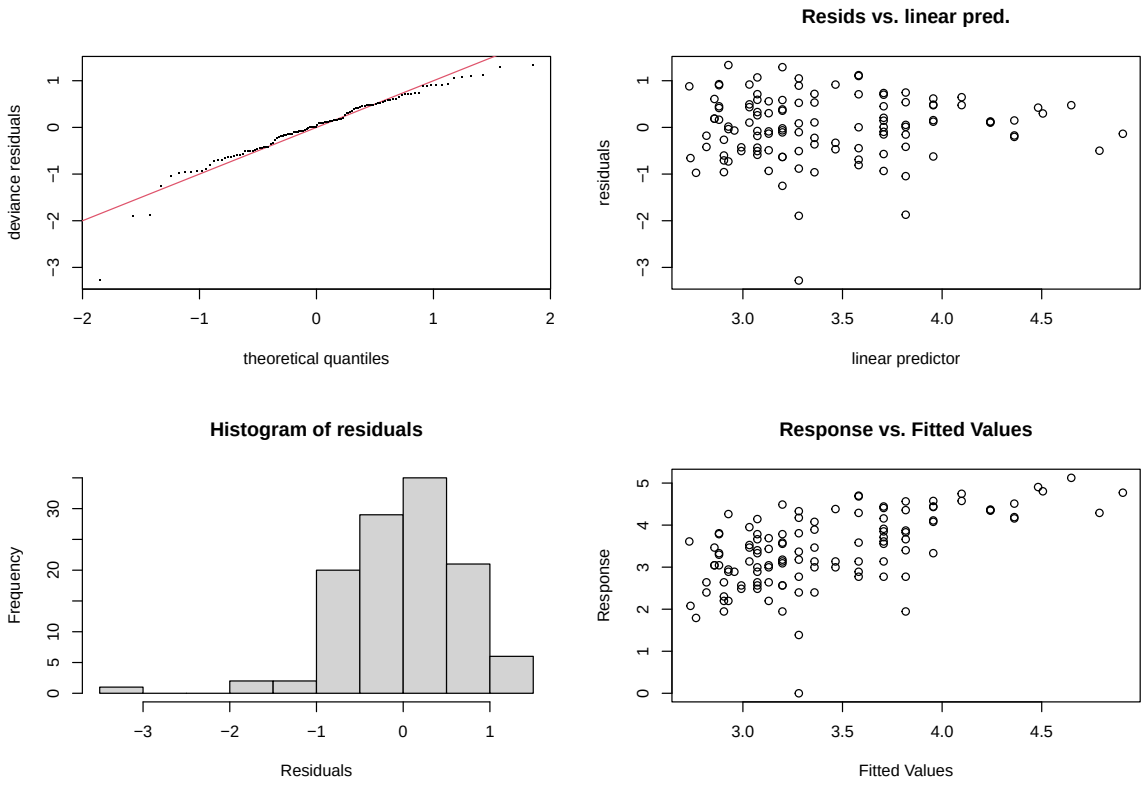



Figure 6: Check plots by ‘gam.check‘

Note that, in model `airq.gam1` we used `k=10` knots, the variable `Wind` has 31 unique values, and hence 10 knots should be enough.

We fit a model for the residuals of the fitted gam model

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## resids ~ s(Wind, k = 20, m = 2)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.137e-15  6.477e-02       0        1
##
## Approximate significance of smooth terms:
##               edf Ref.df F p-value
## s(Wind) 7.419e-05     19 0   0.952
##
## R-sq.(adj) =  -5.66e-07   Deviance explained = 7.89e-06%
## -REML = 124.14  Scale est. = 0.48659   n = 116
```

The results show that there is no unexplained variability between the variable and the residuals. Hence, we can conclude that we do not need more knots. The next Figure supports this statement.

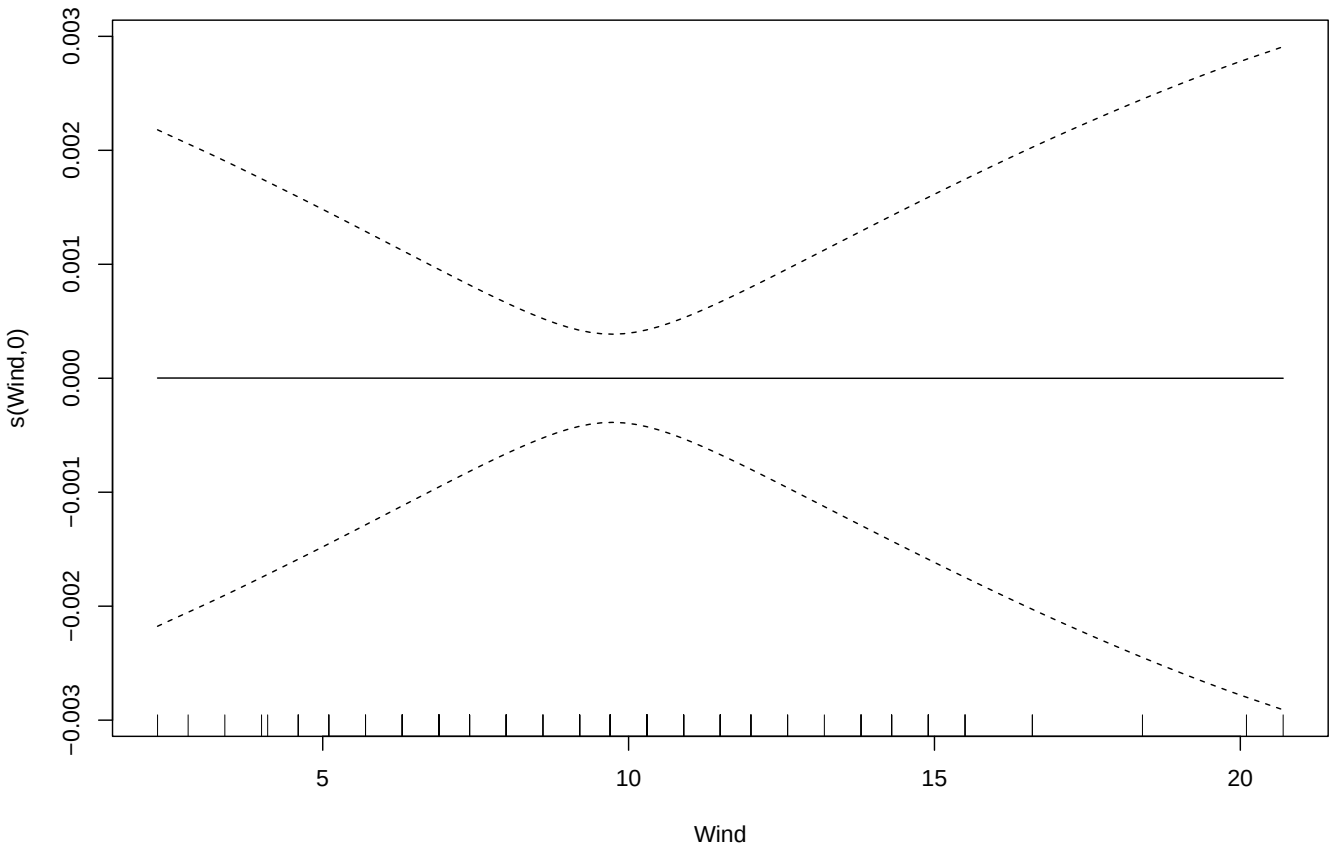



Figure 7: Wind effect vs the residuals of ‘airq.gam1‘

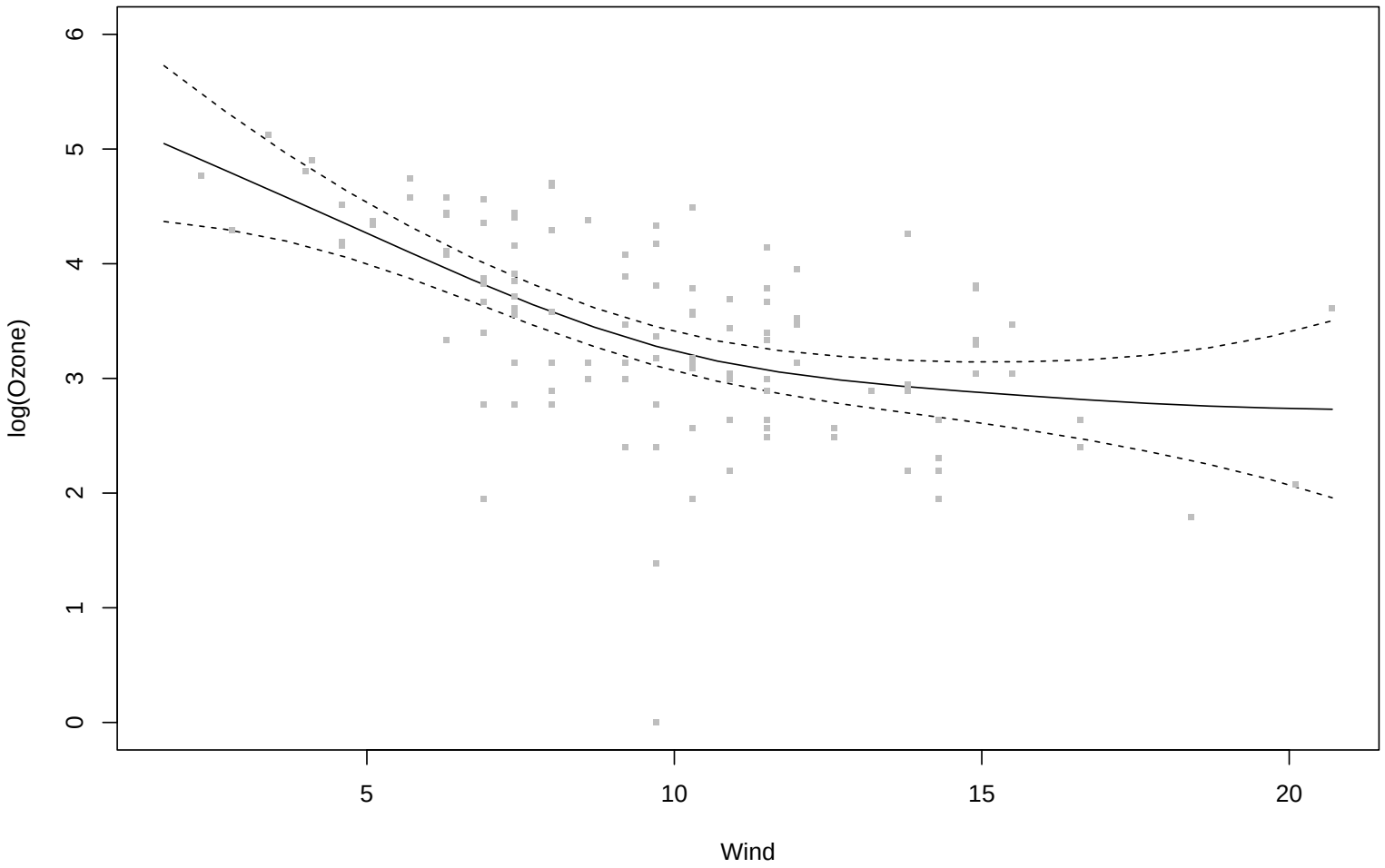



Figure 8: Predicted curve and CI's

Now we add the variable Temp:

```
airq.gam2 <- gam(log(Ozone)~s(Wind,bs="ps",m=2,k=10)+s(Temp,bs="ps",m=2,k=10),
                 method="REML",select=TRUE,data=airquality)
summary(airq.gam2)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## log(Ozone) ~ s(Wind, bs = "ps", m = 2, k = 10) + s(Temp, bs = "ps",
##     m = 2, k = 10)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.41852    0.04963   68.89   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##           edf Ref.df      F p-value
## s(Wind) 2.068      9  1.353 0.00105 **
## s(Temp) 3.990      9 10.496 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.619   Deviance explained = 63.9%
## -REML = 101.64  Scale est. = 0.28568   n = 116
```

Hence, the Temp effect is significative.

The next Figure show both smooth effects.

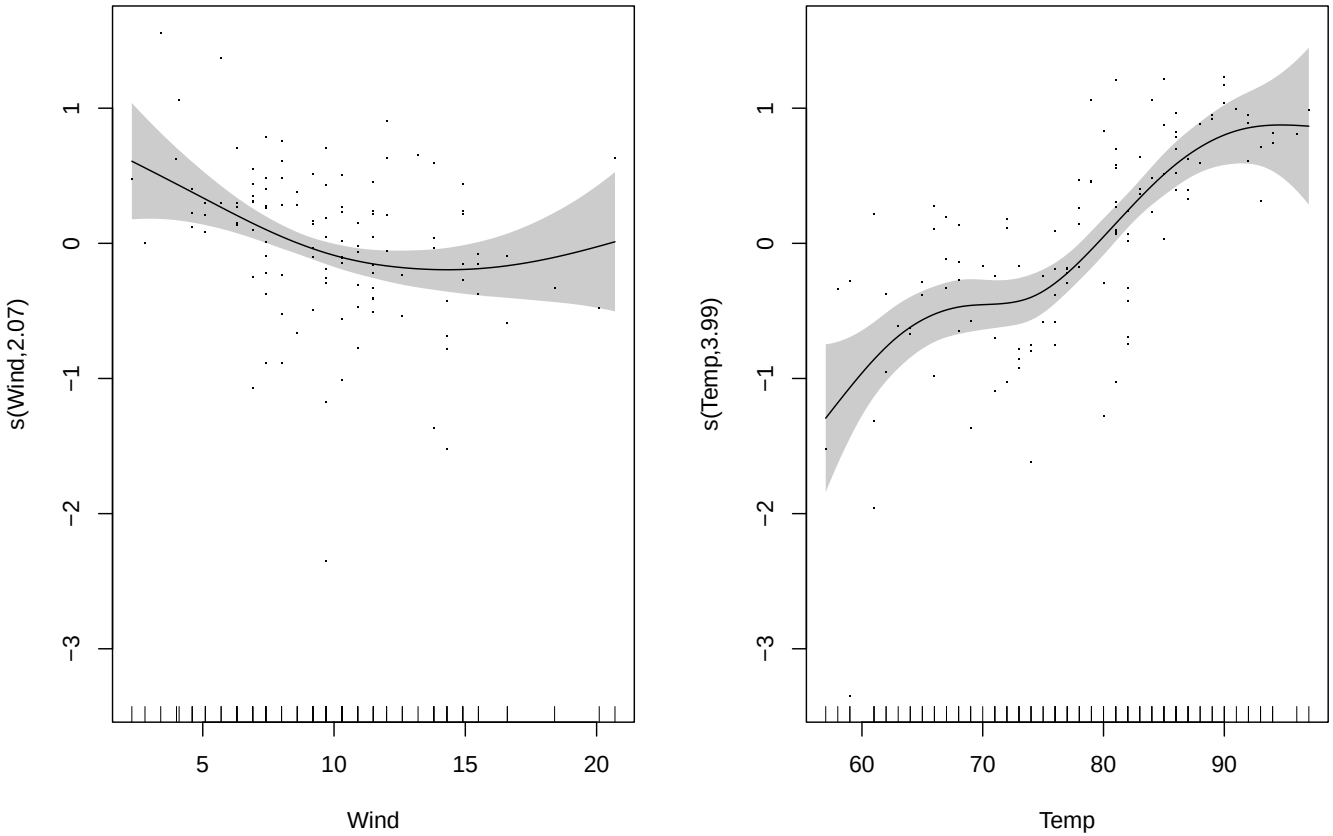



Figure 9: Estimated smooth effect of ‘Wind‘ and ‘Temp‘ on ‘log(Ozone)‘. Data points are the residuals

We can compare both models using the function `anova` for a $F$-test:

```
anova(airq.gam1,airq.gam2)
```

```
## Analysis of Deviance Table
##
## Model 1: log(Ozone) ~ s(Wind, bs = "ps", m = 2, k = 10)
## Model 2: log(Ozone) ~ s(Wind, bs = "ps", m = 2, k = 10) + s(Temp, bs = "ps",
##     m = 2, k = 10)
##   Resid. Df Resid. Dev     Df Deviance
## 1    111.16     55.958
## 2    105.98     31.123 5.1832   24.835
```

We can conclude that `Temp` variable is relevant. Note that, strictly, in this case both models are not nested, as the effective dimension of the variable `Wind` is different when the variable `Temp` is present. Hence, we use the AIC to confirm the results.

```
AIC(airq.gam1)
```

```
## [1] 255.0315
```

```
AIC(airq.gam2)
```

```
## [1] 195.6561
```

Now, we include the covariate `Solar.R`. Notice that this variables contains missing values (`NA`'s).

```
sum(is.na(airquality$Solar.R))
```

```
## [1] 7
```

In order to compare the previous model `airq.gam2` and the new model that includes `Solar.R` we will use the same data, so we will omit the missing values and refit the models.

```
new.airquality <- na.omit(airquality)

airq.gam22=gam(log(Ozone)~s(Wind,bs="ps",m=2,k=10)+s(Temp,bs="ps",m=2,k=10),
               method="REML",select=TRUE,data=new.airquality)

airq.gam3=gam(log(Ozone)~s(Wind,bs="ps",m=2,k=10)+s(Temp,bs="ps",m=2,k=10)+s(Solar.R
              method="REML",select=TRUE,data=new.airquality)
```

```
summary(airq.gam22)
summary(airq.gam3)

AIC(airq.gam22)
AIC(airq.gam3)
```

Is the `Solar.R` variable relevant?

The next Figure shows the estimated smooth effects for `Wind`, `Temp` and `Solar.R` covariables (partial residuals are also plotted).

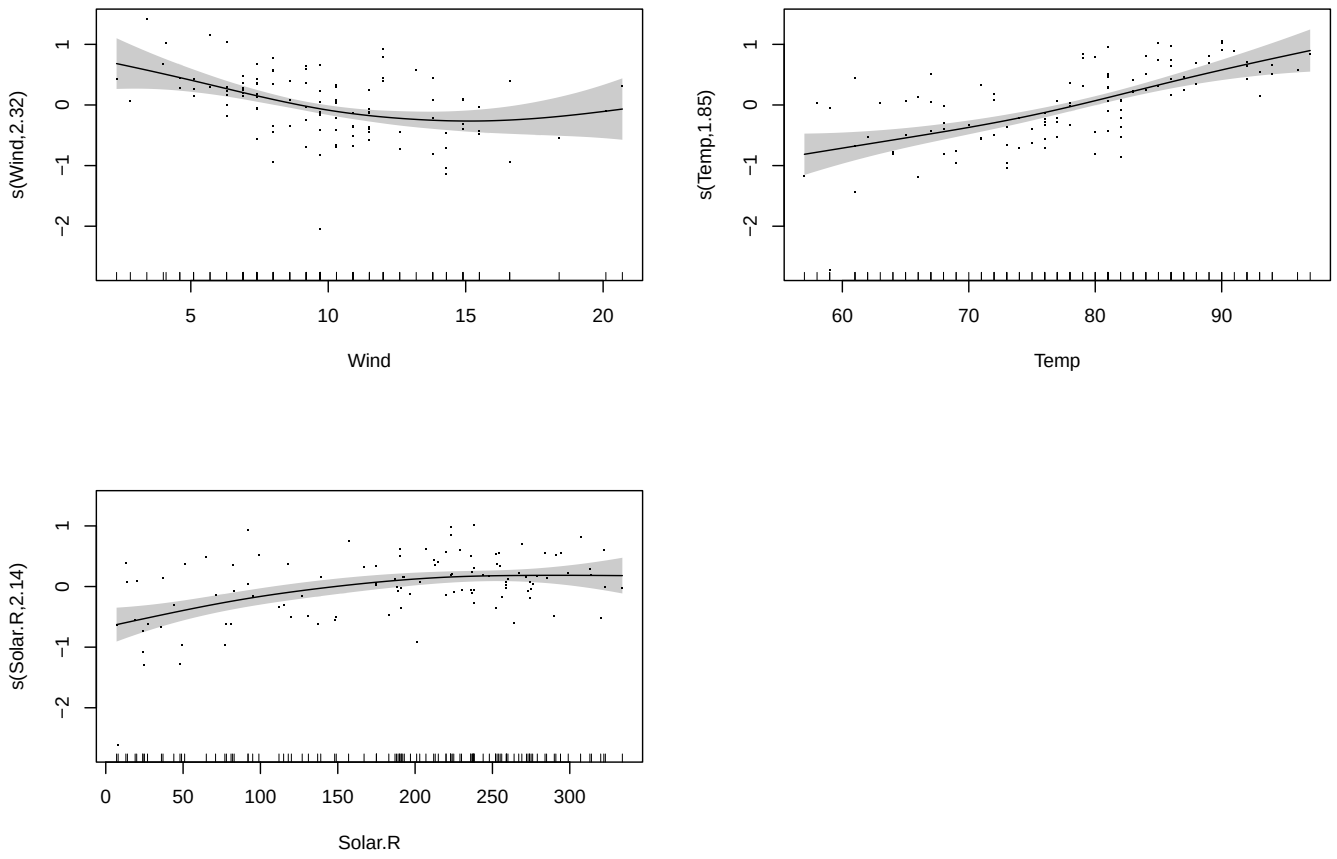



Figure 10: Smooth effects of ‘Wind‘, ‘Temp‘ and ‘Solar.R‘

## Variable selection in GAMs

The function `gamSim` simulates several data for GAMs. See `?gamSim` for details.

```
set.seed(666)
data=gamSim(eg=1,n=400,dist="normal")
fit=gam(y~s(x0)+s(x1)+s(x2)+s(x3),data=data,method="REML",select=TRUE)
summary(fit)
```

The `summary(fit)` table shows the estimated degrees of freedom (`edf`) and the approximated F-test $p$-values. Clearly, the `s(x3)` effect is not significant as we can also see in the estimated effect plot.

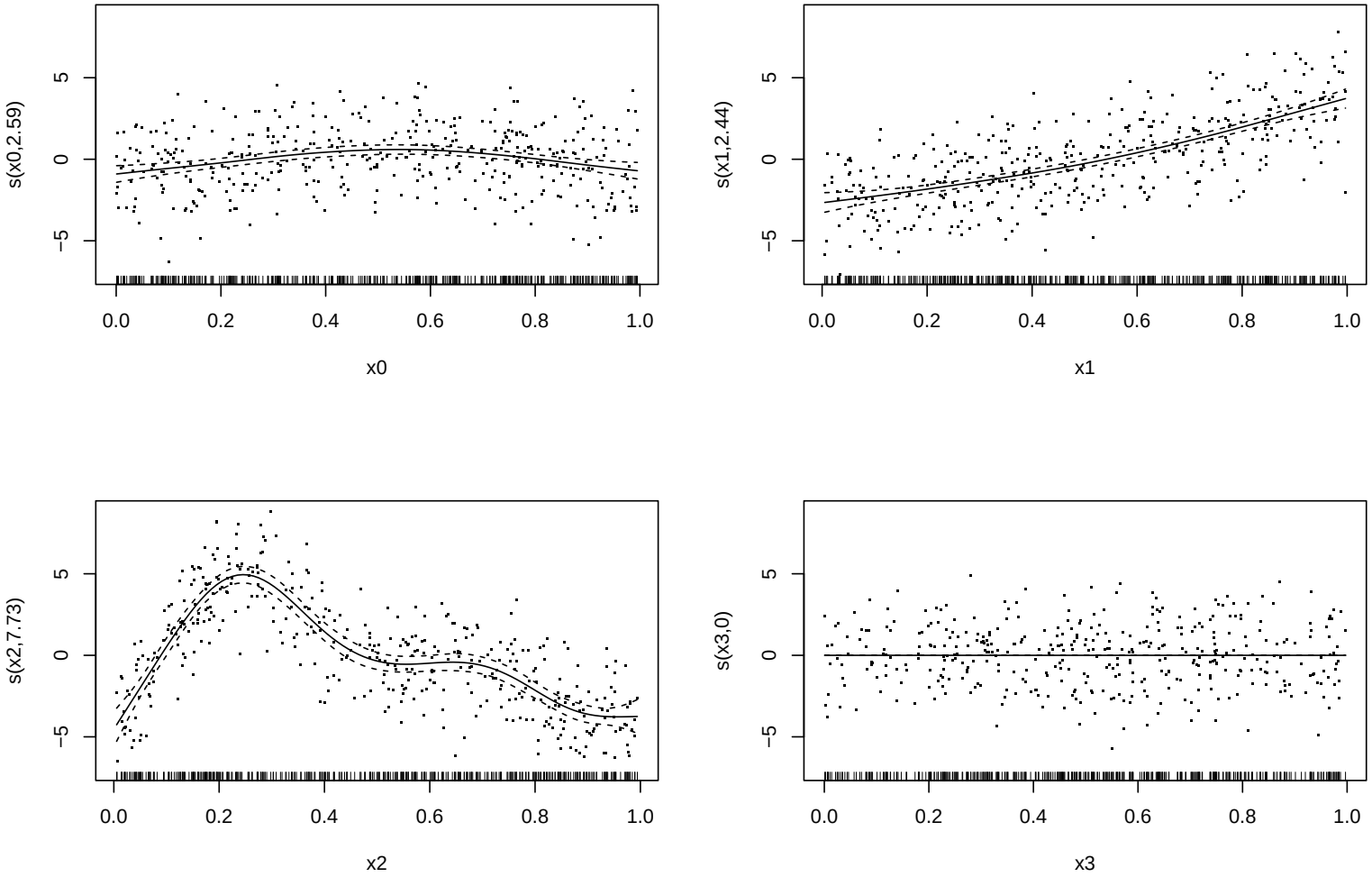



Figure 11: GAM fit with ‘select=TRUE‘

# Semi-parametric models

Semi-parametric models allows mixtures of linear (parametric) and non-parametric components, i.e.:

$$\boldsymbol{y} = \beta_0 + \beta_1 \boldsymbol{x}_1 + ... + \beta_{j-1} \boldsymbol{x}_{j-1} + f(\boldsymbol{x}_j) + \boldsymbol{\epsilon}$$

The fitting procedure is the same as shown in previous sections. Simply the parametric effects are included in the fixed effects part, $\boldsymbol{X}$.

An interested case is then the parametric part includes a factor variable with two or more levels. As we saw in the linear case, we can consider different situations: parallel smooth terms (additive effect) or non-parallel smooth terms (interaction effect).

Moreover, we can consider the same amount of smoothing or different amount.

## Onions data

To illustrate a simple case of a semi-parametric model, we consider the `data(onions)` in the `library(SemiPar)`.
The data consist of 84 observations form an experiment involving the production of white Spanish onions in two South Australian locations.
The variables are:

- `dens` areal density of plants (plants per square metre)
- `yield` onion yield (grammes per plant).
- `location` indicator of location: `0=Purnong Landing`, `1=Virginia`.

The next Figure shows that for higher density of plants the Purnong Landing yield of onios is greater.

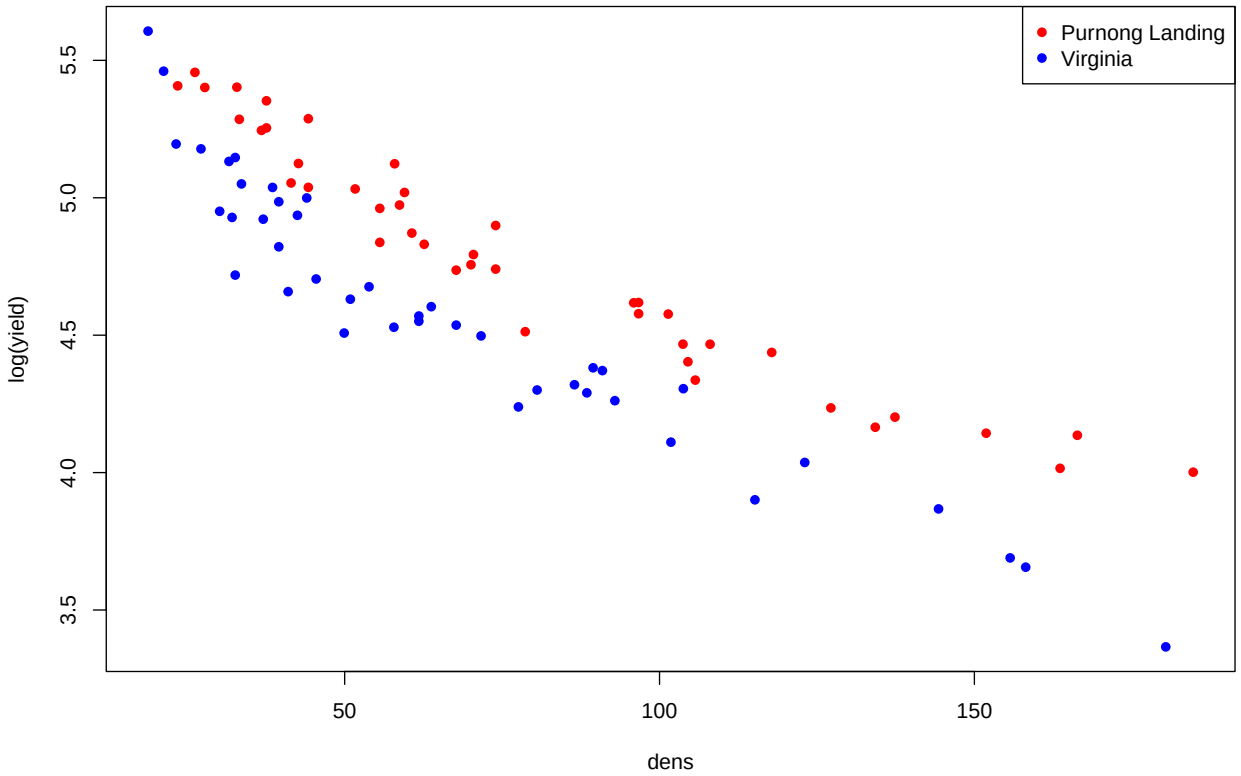



Figure 12: Onions yield in two locations

The linear model will be

$$\log(\text{yield}_i) = \beta_0 + \beta_1 \text{location}_{ij} + \beta_2 \text{dens}_i + \epsilon_i$$

where

$$\text{location}_{ij} = \begin{cases} 0 & \text{if the } i\text{th observation is from Purnong Landing} \\ 1 & \text{if the } i\text{th observation is from Virginia} \end{cases}$$

```r
# create a factor for location
L <- factor(location)

levels(L) <- c("Purnong Landing","Virginia")

# fit a gam with location factor
fit1 <- gam(log(yield) ~ L + s(dens,k=20,m=2,bs="ps"),
            method="REML", select=TRUE)
```

```r
summary(fit1)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## log(yield) ~ L + s(dens, k = 20, m = 2, bs = "ps")
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.85011    0.01688  287.39   <2e-16 ***
## LVirginia   -0.33284    0.02409  -13.82   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##           edf Ref.df     F p-value
## s(dens) 4.568     19 72.76  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.946   Deviance explained = 94.9%
## -REML = -54.242  Scale est. = 0.011737  n = 84
```

The next Figure shows the fitted curves for each location with model `fit1`, we assumed both curves are parallel.

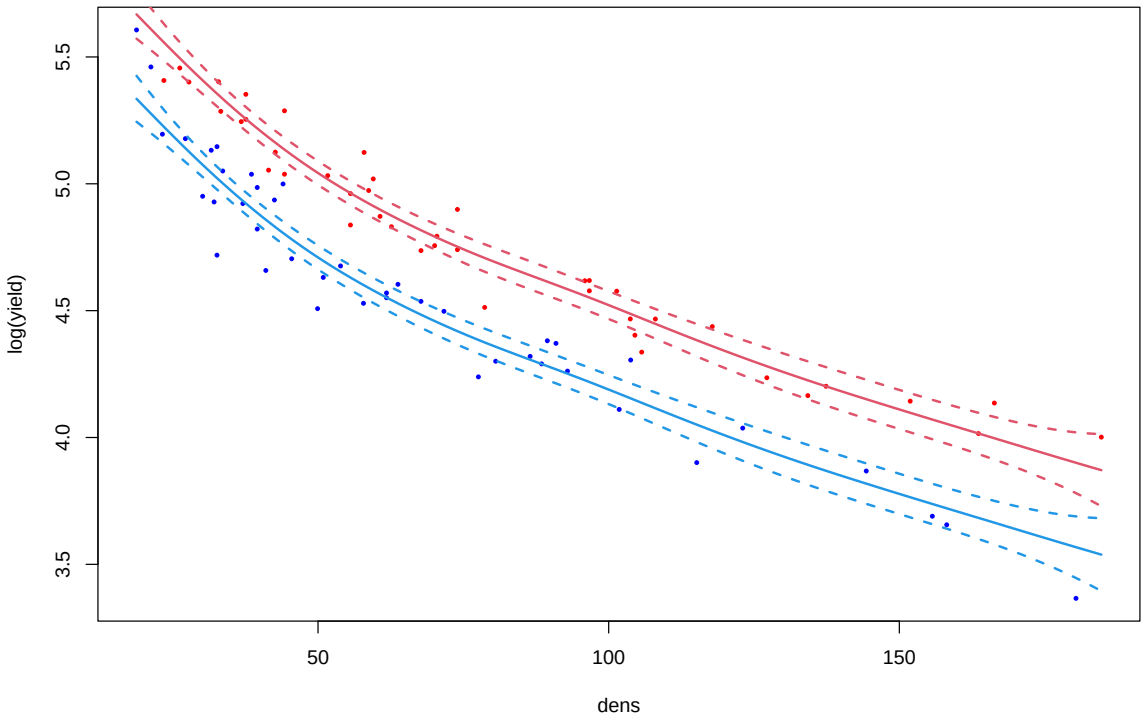



Figure 13: Fitted curves for each location

Assuming parallel curves for both locations implies that the decrease of the yield as the density increases is the same in both locations. Instead of fitting an additive effect model we can fit an interaction effect model such as:

$$\log(\text{yield}_i) = \beta_0 + \beta_1\beta_1\text{location}_{ij} + f(\text{dens}_i)_{L(j)} + \epsilon_i$$

where

$$L(j) = \begin{cases} 0 & \text{if the } i\text{th observation is from Purnong Landing} \\ 1 & \text{if the } i\text{th observation is from Virginia} \end{cases}$$

```r
fit2 <- gam(log(yield) ~ L + s(dens,k=20,m=2,bs="ps",by=L),
            method="REML", select=TRUE)
summary(fit2)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## log(yield) ~ L + s(dens, k = 20, m = 2, bs = "ps", by = L)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.84407    0.01603  302.12   <2e-16 ***
## LVirginia   -0.33003    0.02271  -14.54   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                            edf Ref.df     F p-value
## s(dens):LPurnong Landing 3.097     18 37.62  <2e-16 ***
## s(dens):LVirginia        4.728     17 52.10  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.952   Deviance explained = 95.7%
## -REML = -53.616  Scale est. = 0.01045   n = 84
```

## Which model is better?

```r
AIC(fit1)
```

```
## [1] -125.2307
```

```r
AIC(fit2)
```

```
## [1] -131.1844
```

```r
fit1$sp
```

```
##     s(dens)1     s(dens)2
## 164.66124239   0.00147493
```

```r
fit2$sp
```

```
## s(dens):LPurnong Landing1 s(dens):LPurnong Landing2        s(dens):LVirginia1
##              4.264039e+02              1.604648e-03              6.437458e+01
##        s(dens):LVirginia2
##              1.053443e-03
```

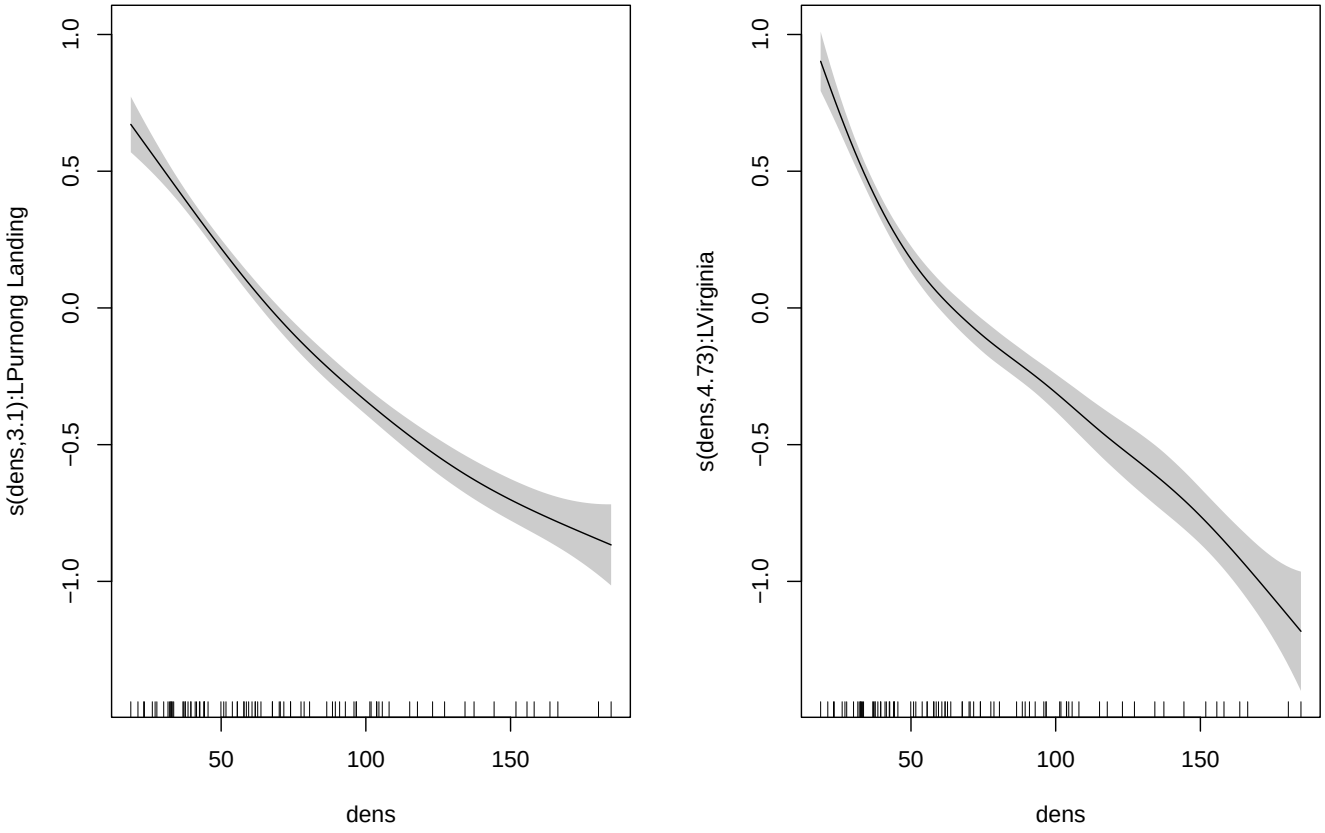



Figure 14: plot.gam

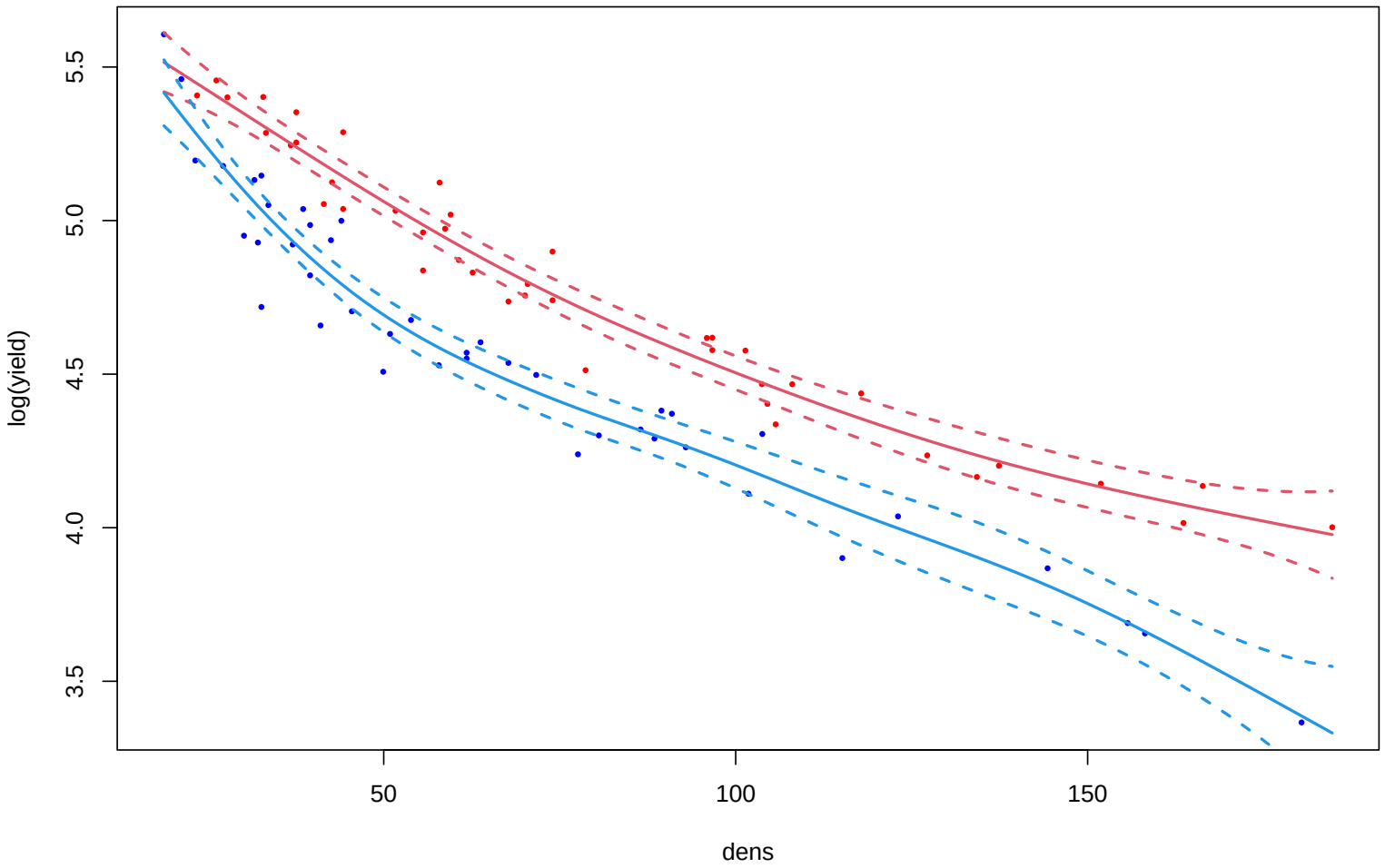



Figure 15: Fitted curves with interaction

# Penalized splines for longitudinal data

In this section we will consider longitudinal data in which the individuals trajectories are modelled as a non-linear function of time.

## Girls leukemia data

We will use data from a longitudinal study related to different therapies on girls. The data where slightly modified to preserve individuals privacy.
The variables are

- `case` individual
- `treatment` type of treatment received (`1=no radiation`, `2=conventional radiation` or `3=hyperfractioned radiation`)
- `height` girl height in cm
- `age` age in years

The total number of observations is 1988. The number of measurement lies between 1 and 21.

**Random intercept model:**

The simplest model is of the form:

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \sum_{l=2}^{L} \gamma_l \mathsf{tr}_{il} + U_i + \epsilon_{ij}$$

$$U_i \sim N(0, \sigma_U^2) \quad \epsilon_{ij} \sim N(0, \sigma_\epsilon) \quad \text{for} \quad \begin{array}{l} 1 \leq i \leq 197 \\ 1 \leq j \leq n_i \end{array}$$

where

$$\mathsf{tr}_{il} = \begin{cases} 1 \text{ if the } i\text{th girl received treatment } l \\ 0 \text{ otherwise} \end{cases}$$

This model assumes that all girls have the same linear growth rate and the variability among girls are accounted by the random effect $U_i$. In matrix form:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}\boldsymbol{u} + \boldsymbol{\epsilon} \tag{4.1}$$

where

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_1 & \boldsymbol{T}_1 \\ \vdots & \vdots \\ \boldsymbol{X}_m & \boldsymbol{T}_m \end{pmatrix}, \ \boldsymbol{X}_i = \begin{pmatrix} 1 & x_{i1} \\ \vdots & \vdots \\ 1 & x_{in_i} \end{pmatrix}, \ \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{pmatrix}, \ \boldsymbol{Z} = \begin{pmatrix} \mathbf{1}_1 & 0 & \dots & 0 \\ 0 & \mathbf{1}_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathbf{1}_m \end{pmatrix},$$

and

$$\mathbf{1}_i = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}_{n_i \times 1}$$

$\boldsymbol{T}_i$ is a matrix that indicates if the $i$th girls receives treatment $1, 2$ or $3$.

In R

```
library(gamlss)
data(Leukemia)
```

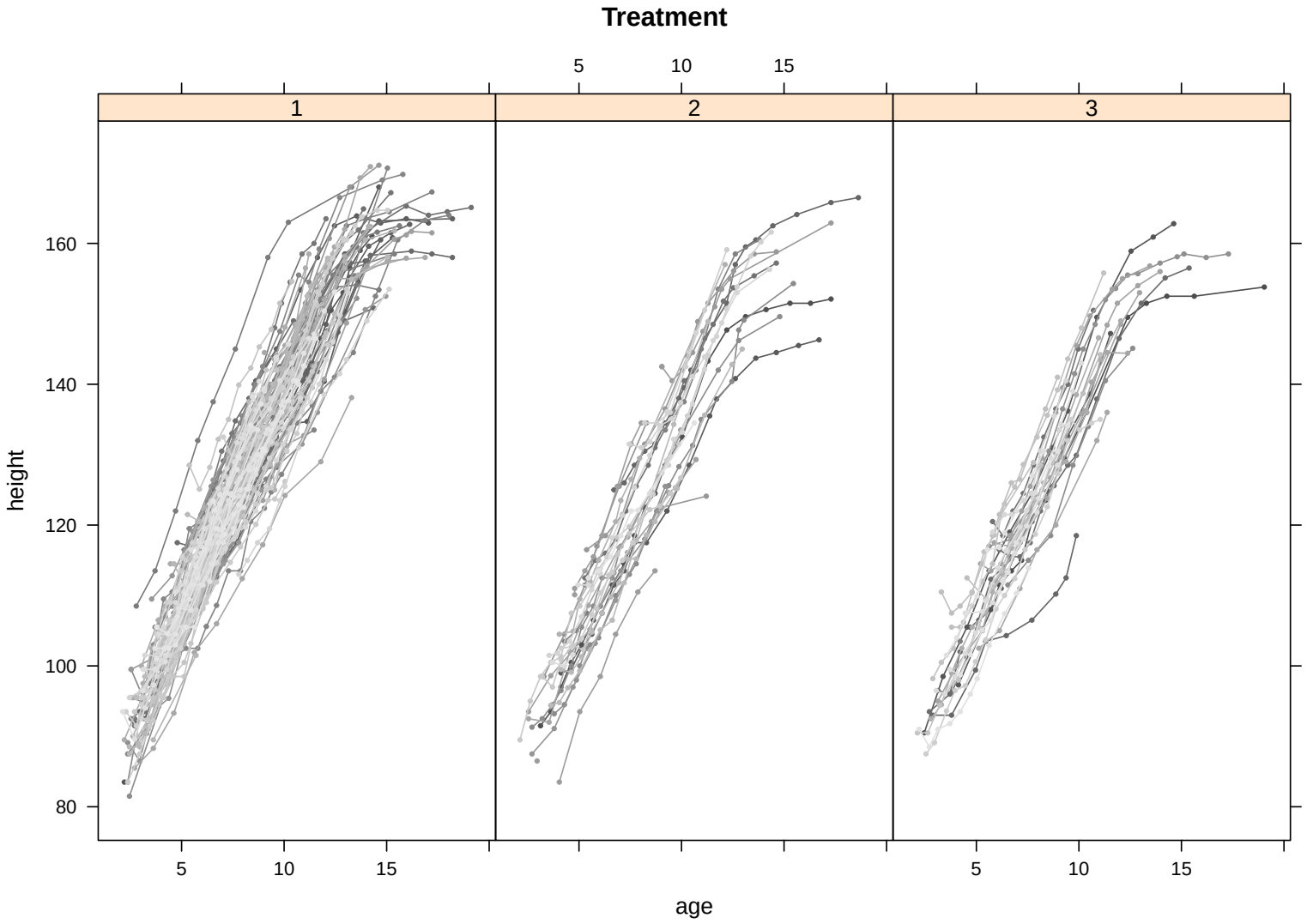



Figure 16: Girls data

## Random intercept model

The next Figure shows the fitted straight lines for the random intercept model. This model is not able to describe the trajectories of the girls' height.

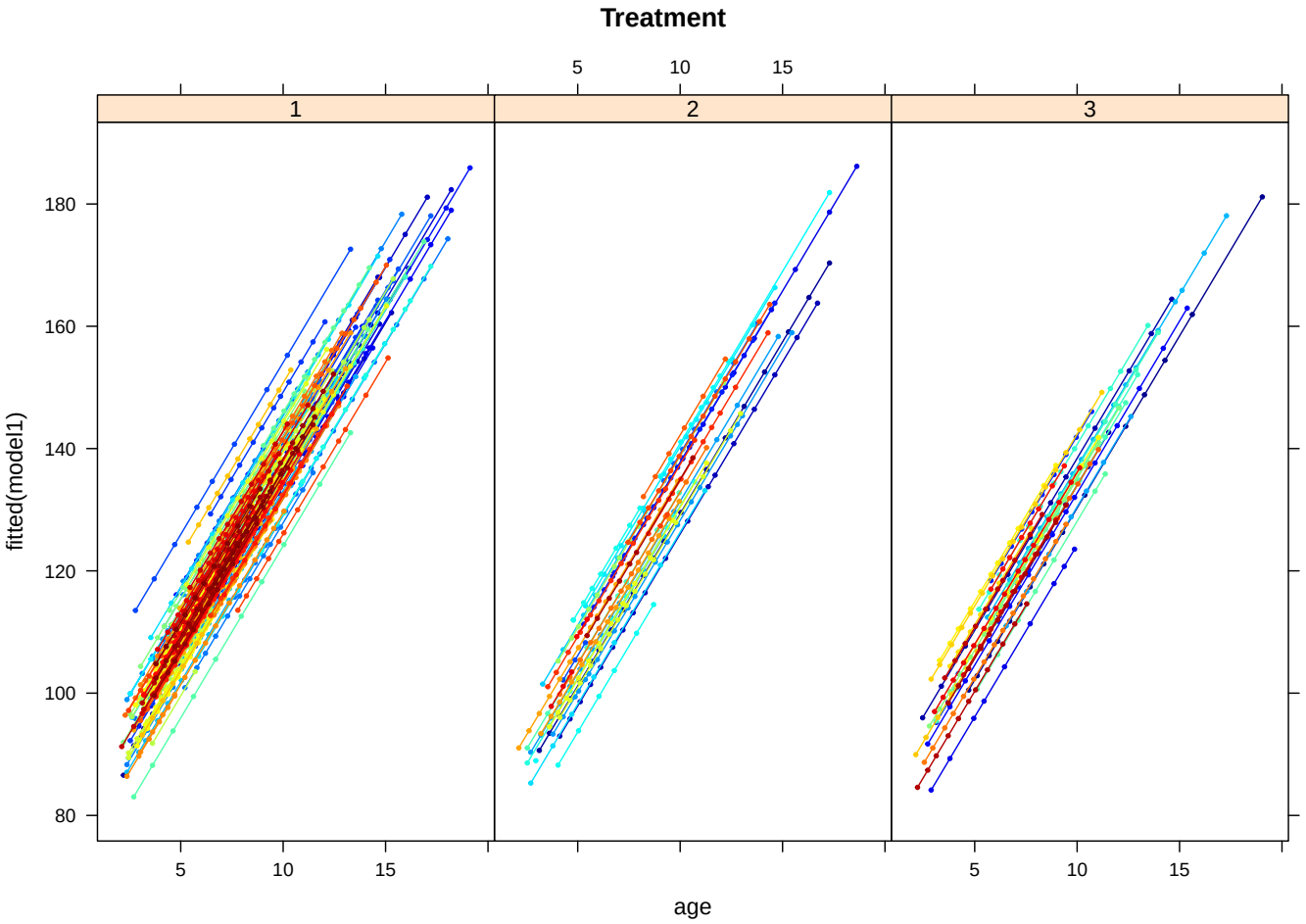



Figure 17: Random intercept fitted model per treatments

**Additive mixed model** A natural extension of the previous model is

$$y_{ij} = \sum_{l=2}^{L} \gamma_l \mathsf{tr}_{il} + f(x_{ij}) + U_i + \epsilon_{ij} \quad U_i \sim N(0, \sigma_U^2) \quad \epsilon_{ij} \sim N(0, \sigma_\epsilon)$$

where $f$ is a smooth function accounting for the growth trend of the girls. The function $f$ is estimated with $P$-splines (using the mixed model representation). Now,

$$\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{Z}_1 & \mathbf{1}_1 & 0 & \dots & 0 \\ \boldsymbol{Z}_2 & 0 & \mathbf{1}_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{Z}_m & \vdots & \vdots & \dots & \mathbf{1}_m \end{pmatrix}$$

where $\boldsymbol{Z}_i$ is the random effect matrix obtained from the mixed model representation of a $P$-spline.

$$\boldsymbol{u} = (u_1, u_2, ...., u_k, U_m)' \quad \text{and} \quad \boldsymbol{G} = \mathsf{Cov}(\boldsymbol{u}) = \begin{pmatrix} \sigma_u^2 \boldsymbol{I} & 0 \\ 0 & \sigma_U^2 \boldsymbol{I} \end{pmatrix}$$

Instead of straight lines, we fit smooth curves that only differs in their intercepts.

# In R we use the function gamm

```
library(mgcv)
# Smooth random intercept model
fit2.gamm <- gamm(height~factor(treatment)+s(age,k=40,bs="ps",m=2),
                  random=list(case=pdIdent(~1)),data=Leukemia)
```

The smooth age effect is shown in the next figure.

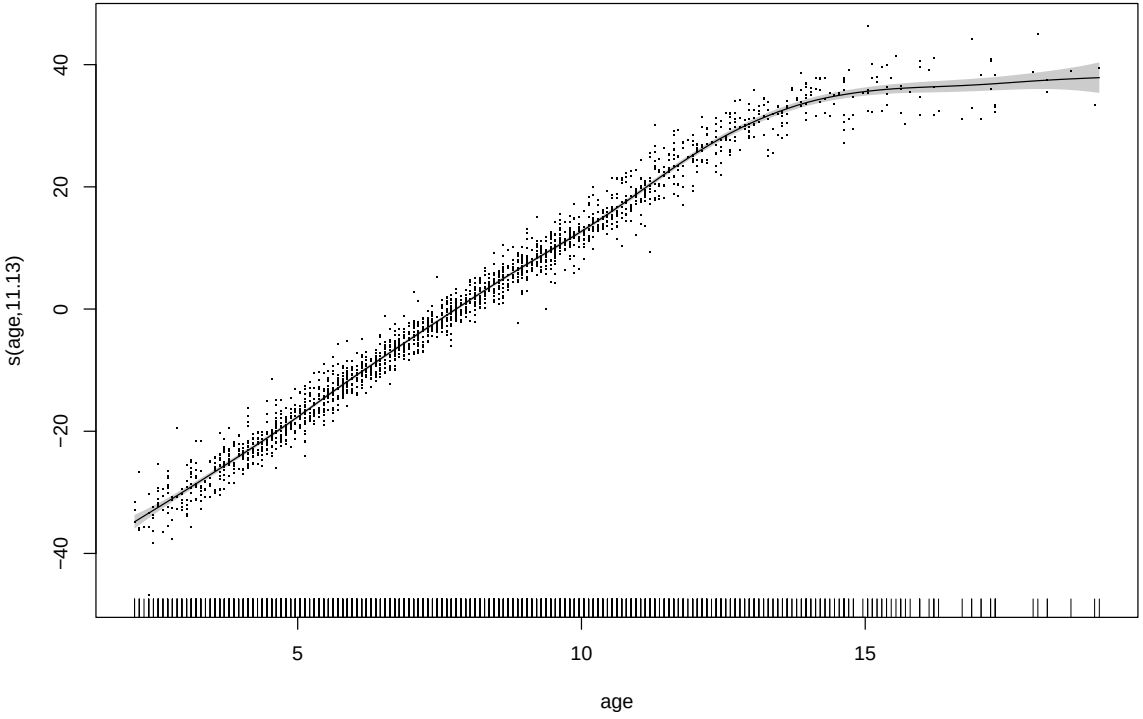



Figure 18: Smooth effect for the random intercept model

We can plot the fitted curves using the function `xyplot` from the library `lattice`, in order to plot the fitted curves and overlap the original data points we will use the library `latticeExtra` and the function `as.layer`.

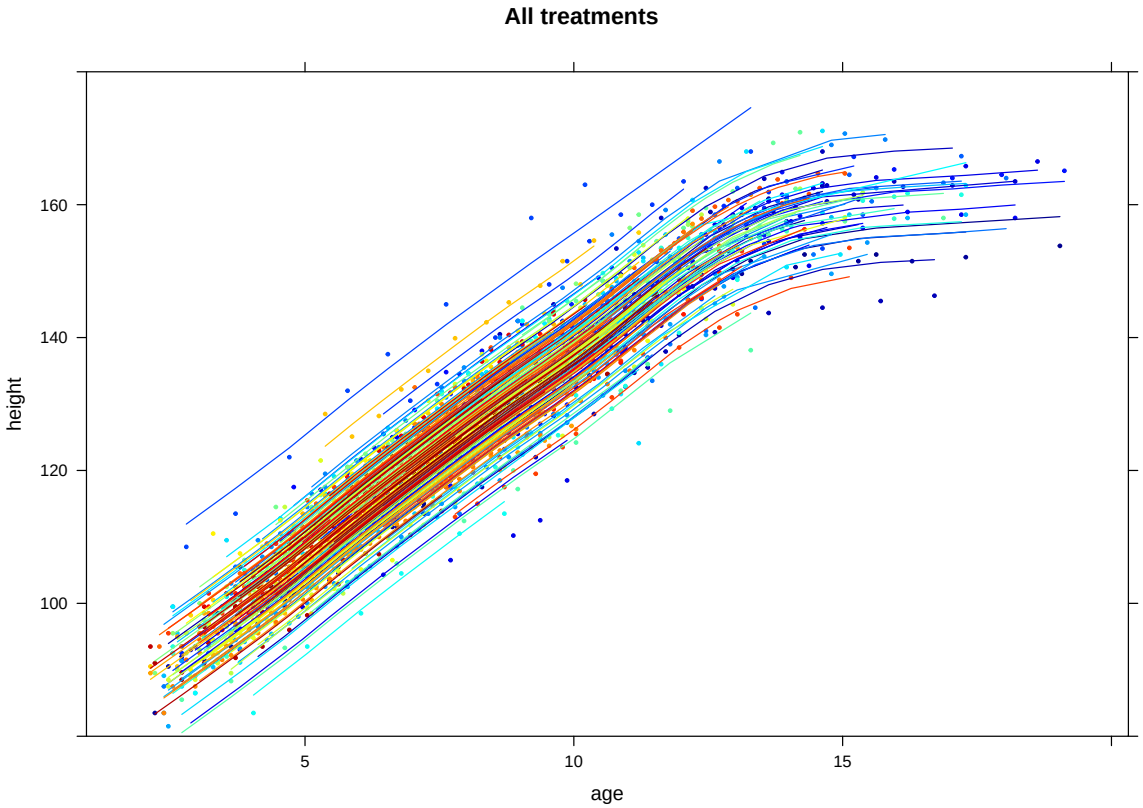



Figure 19: Smooth effect for the random intercept model

The next plot shows the fitted trends by treatment.

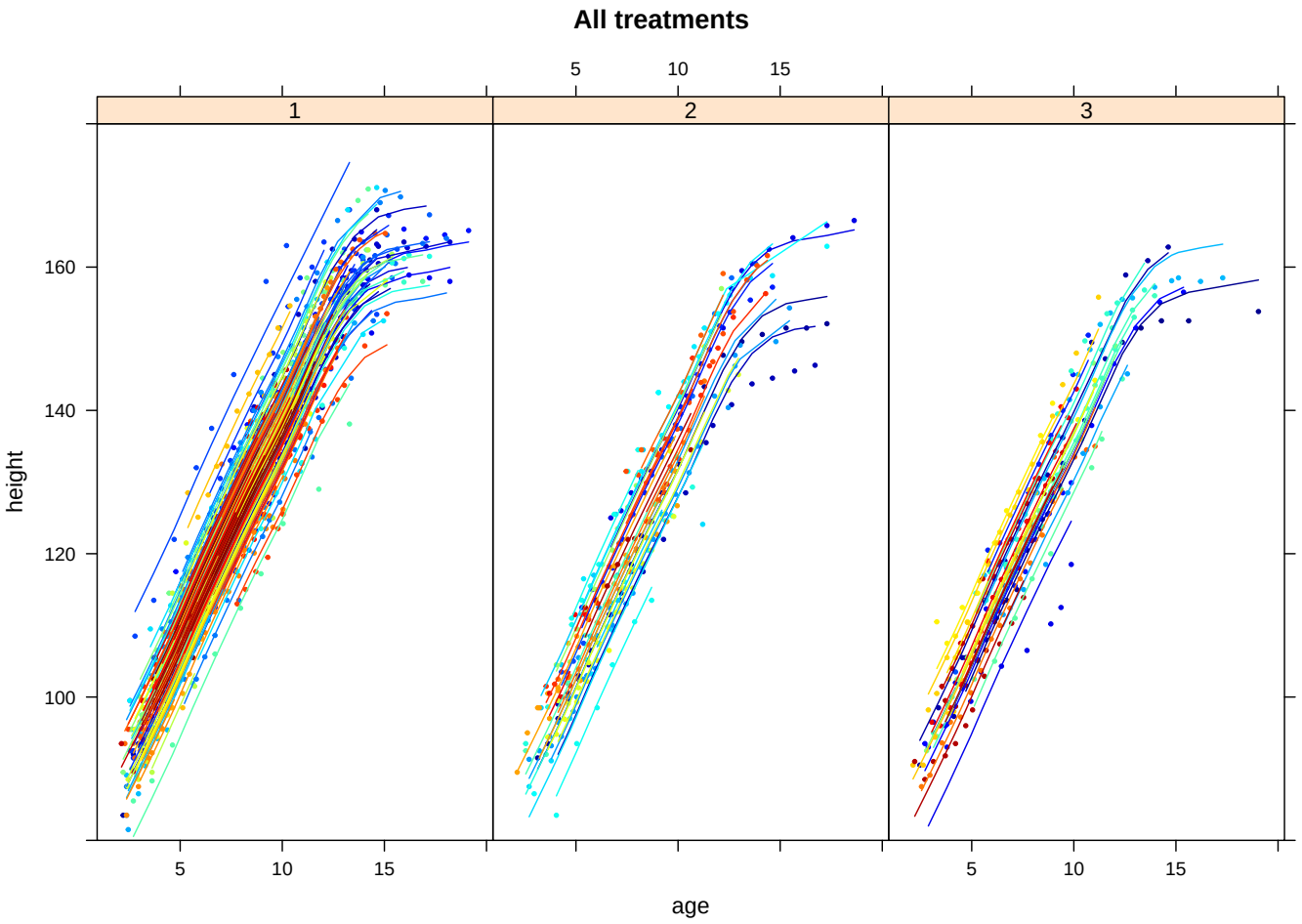



Figure 20: Fitted trends for the smooth random intercept model by treatment

Previous plots show that we account for the non-linear trend but we are not able to capture the individual trajectories.

**Linear individual differences model (Random intercept and slope model)**

This model is an extension of the previous one. Now we allow not only for random intercept but also for random slopes.

$$y_{ij} = \sum_{l=2}^{L} \gamma_l \mathsf{tr}_{il} + f(x_{ij}) + a_{i1} + a_{i2} x_{ij} + \epsilon_{ij} \quad \epsilon_{ij} \sim N(0, \sigma_\epsilon) \qquad (a_{i1}, a_{i2})' \sim N(0, \Sigma)$$

In matrix notation, now the random effect matrix $\boldsymbol{Z}$ is

$$\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{Z}_1 & \boldsymbol{X}_1 & \boldsymbol{0} & \dots & \boldsymbol{0} \\ \boldsymbol{Z}_2 & \boldsymbol{0} & \boldsymbol{X}_2 & \dots & \boldsymbol{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{Z}_m & \boldsymbol{0} & \boldsymbol{0} & \dots & \boldsymbol{X}_m \end{pmatrix},$$

with random effects:

$$\boldsymbol{u} = (u_1, \dots, u_K, a_{11}, a_{12}, \dots, a_{m_1}, a_{m_2})'$$

and covariance matrix:

$$\boldsymbol{G} = \mathsf{Cov}(\boldsymbol{u}) = \begin{pmatrix} \sigma_u^2 \boldsymbol{I} & 0 \\ 0 & \underset{1 \le i \le m}{\mathsf{blockdiagonal}} \boldsymbol{\Sigma} \end{pmatrix}$$

```r
# Random intercepts and slopes and smooth effect for age
fit3.gamm <- gamm(height~factor(treatment)+s(age,k=40,bs="ps",m=2),
                  random=list(case=pdSymm(~age)),data=Leukemia)
summary(fit3.gamm$gam)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## height ~ factor(treatment) + s(age, k = 40, bs = "ps", m = 2)
##
## Parametric coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        125.5430     0.4507 278.562   <2e-16 ***
## factor(treatment)2  -2.2864     0.9637  -2.373   0.0178 *
## factor(treatment)3  -1.3464     0.9734  -1.383   0.1668
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##          edf Ref.df    F p-value
## s(age) 12.05  12.05 1198  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.919
##   Scale est. = 3.6616    n = 1988
```

This model adds more flexibility, allowing for non-parallel curves.

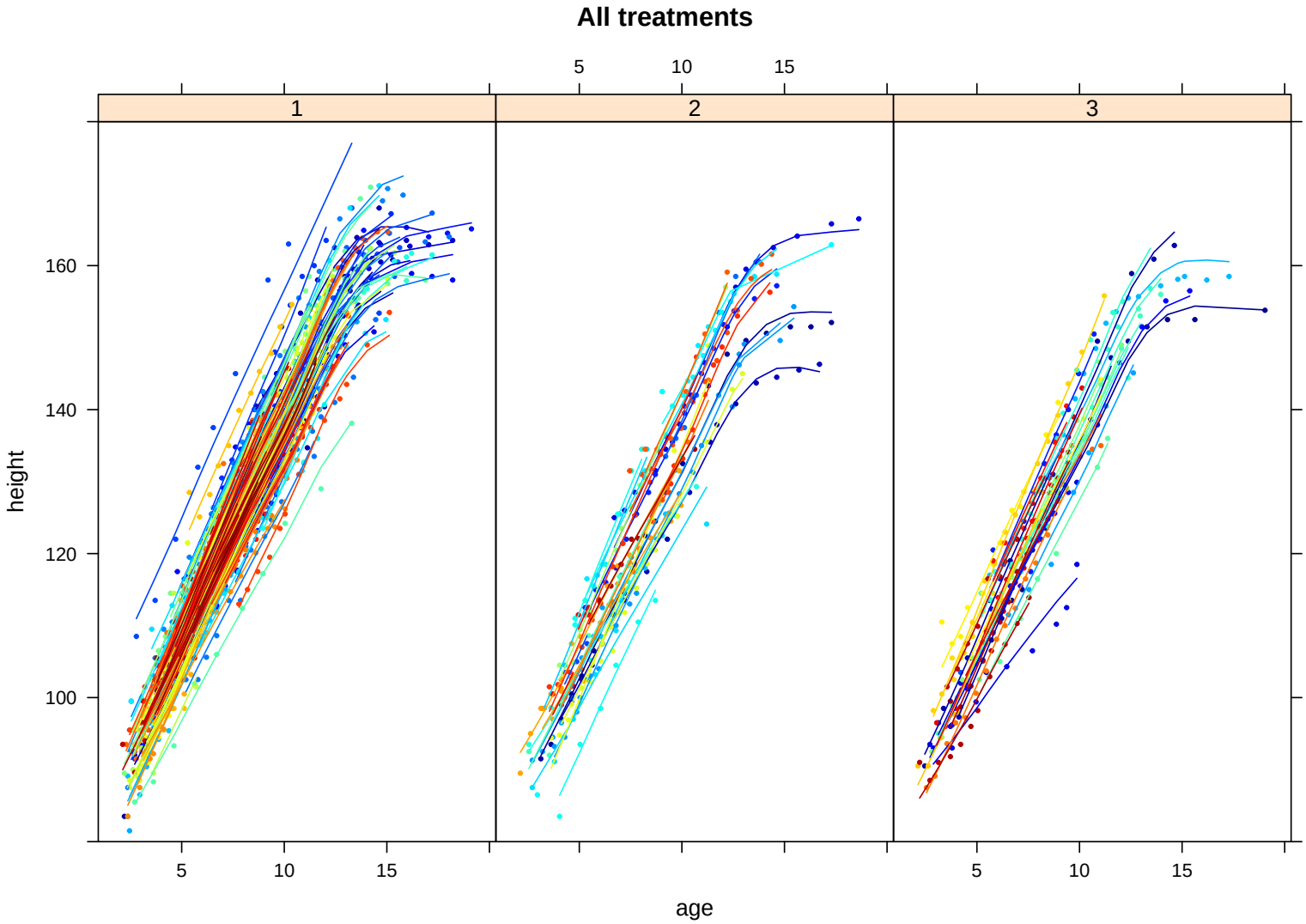



Figure 21: Fitted individual trends for the smooth random intercept and slope model by treatment

## Curve by factor interaction

The final aim of the study was the long term effects of the three different therapies. Hence, individual curves for each treatment is of interest. We can extend the model by allowing for a interaction of a factor variable with a continuous predictor. The model is of the form:

$$y_{ij} = f_{z_i}(x_{ij}) + a_{i1} + a_{i2}x_{ij} + \varepsilon_{ij}$$

more formally

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + Z_i u_k \sum_{l=2}^{L} \mathsf{tr}_{il}(\gamma_{0l} + \gamma_{1l}x_{ij}) + \sum_{l=2}^{L} \mathsf{tr}_{il} Z_i w_k^l + a_{i1} + a_{i2}x_{ij} + \varepsilon_{ij}$$

where

$$w_k^l \sim N(0, \sigma_{wl}^2), \quad (a_{i1}, a_{i2})' \sim N(0, \Sigma) \quad , \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2),$$

where $\mathsf{tr}_{il} = 1$ if $\mathsf{tr}_i = l$ and $0$ otherwise.

In a model with factor variables, we must impose a constraint to ensure an identifiable model, in this case, we impose that $\gamma_{01} = \gamma_{11} = 0$, which means that $\beta_0 + \beta_1 x_{ij} + Z_i u_k$ is the fitted curve for $l = 1$, and $\gamma_{0l} + \gamma_{1l}x_{ij} + Z_i w_k^l$ is the difference between the fitted curves for therapy 2 (conventional radiation) and therapy 3 (hyperfractioned radiation) and therapy 1 (no radiation).

```r
# Curve by treatment model
fit4.gamm <- gamm(height~factor(treatment)+s(age,k=40,bs="ps",m=2,
                                             by=factor(treatment)),
                  random=list(case=pdSymm(~age)),data=Leukemia)
summary(fit4.gamm$gam)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## height ~ factor(treatment) + s(age, k = 40, bs = "ps", m = 2,
##     by = factor(treatment))
##
## Parametric coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        125.8952     0.4606 273.358  < 2e-16 ***
## factor(treatment)2  -3.6383     1.0508  -3.463 0.000547 ***
## factor(treatment)3  -2.2342     1.0605  -2.107 0.035271 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                              edf Ref.df     F p-value
## s(age):factor(treatment)1 11.589 11.589 954.8  <2e-16 ***
## s(age):factor(treatment)2  8.986  8.986 250.8  <2e-16 ***
## s(age):factor(treatment)3  9.251  9.251 268.3  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.918
##   Scale est. = 3.41      n = 1988
```

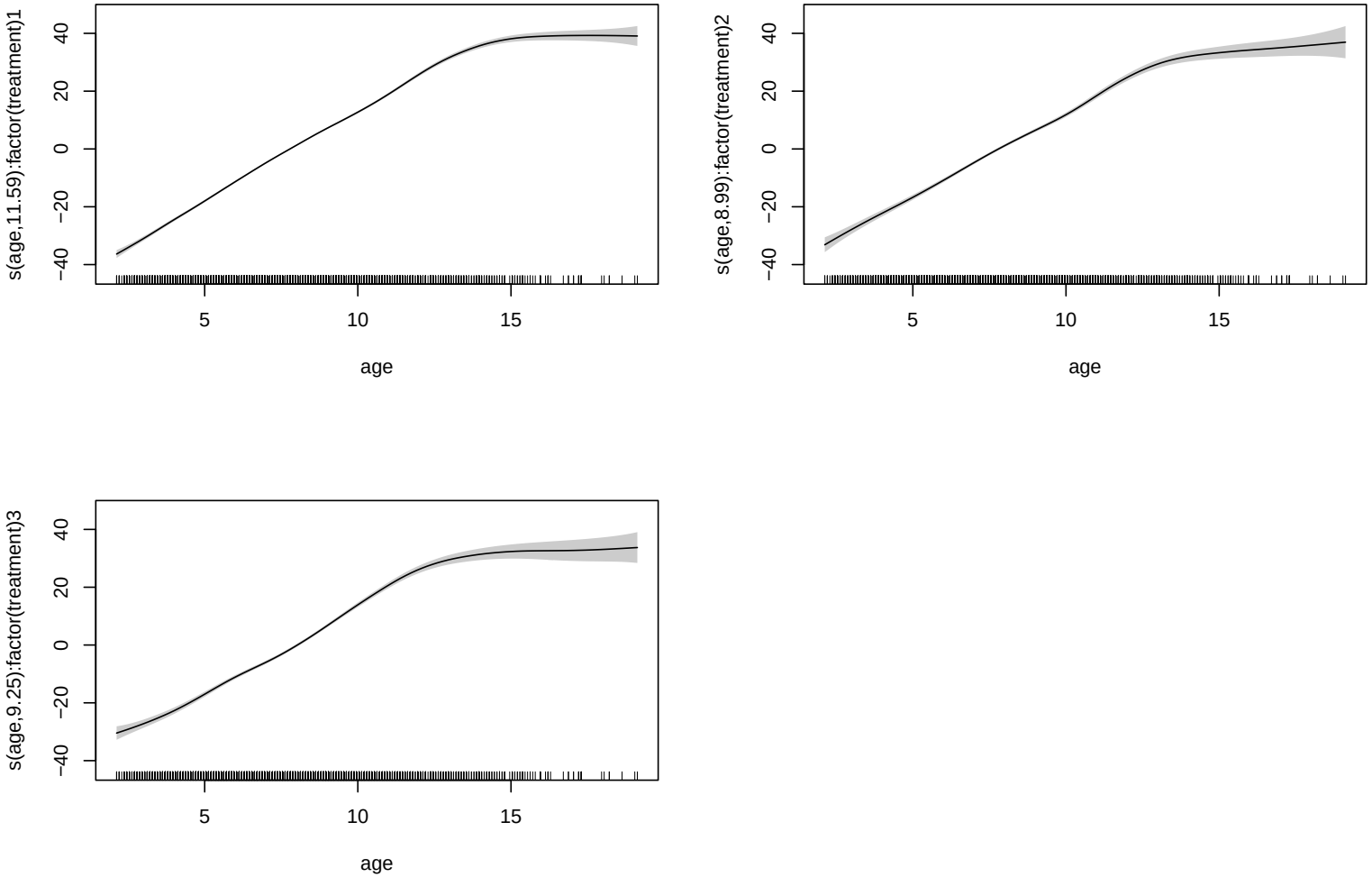



Figure 22: Fitted global trends for the smooth random intercept and slope model by treatment

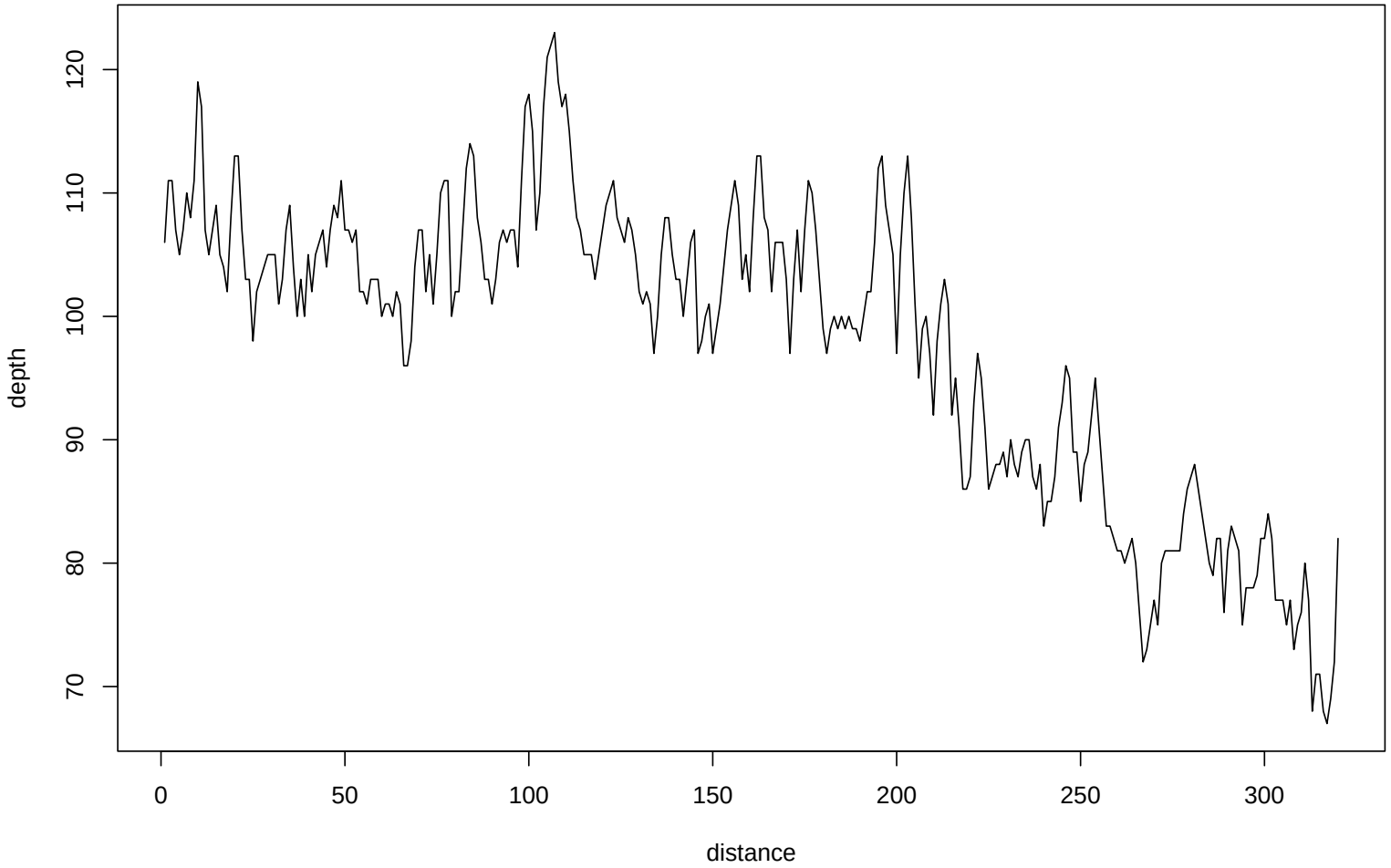
# P-splines for correlated data

The data were analyzed in Pandit and Wu (1983). They present a dataset (`wood.txt`) describing 320 measurements of a block of wood that was subject to grinding.

```
wood<-read.table("https://idaejin.github.io/courses/UPC2022/data/wood.txt",header=TR
```

Next Figure shows the profile (depth) height at different distances. The profile variation follows a curve determined by the radius of the grinding stone.

```r
with(wood,plot(distance,depth,t="l"))
```



** Try to fit the data with `smooth.spline` **

Using $P$-splines we can estimate a smooth trend and possible correlation simultaneously. The mixed model representation of $P$-splines helps us to estimate both effects pretty easily. Let us consider, firstly a smooth model that ignores the correlation between the observations.

The left figure shows the estimated smooth trend. This shows that ignoring the correlation gives a non-smooth trend. The right panel show the autocorrelation function with correlation in the errors.

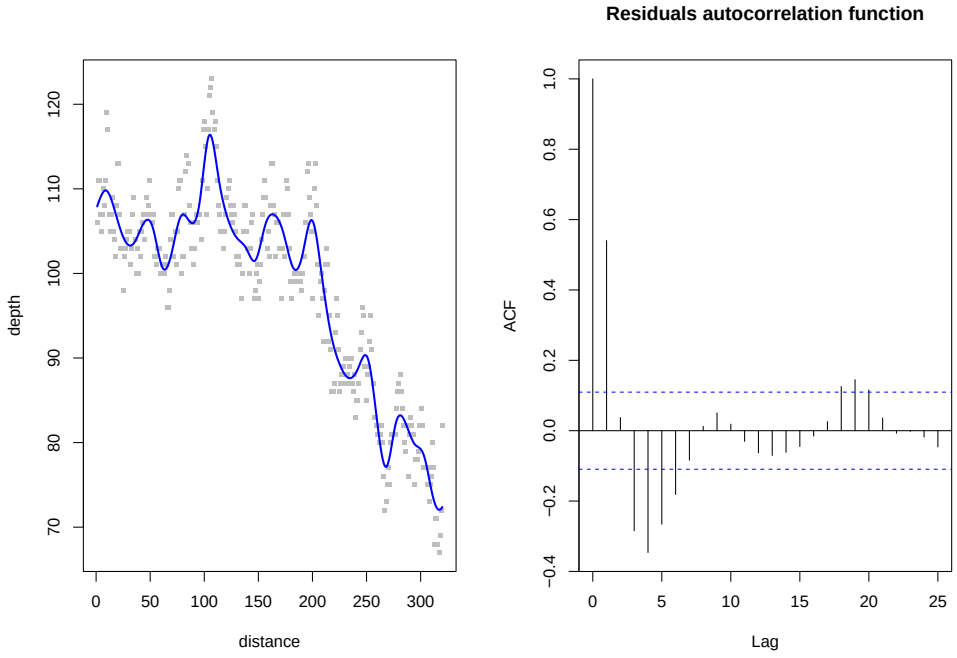



Figure 23: Estimated trend and residuals autocorrelation

A correlated data model can be added with an AR(1) or AR(2) structure, i.e.

```
# create a factor per observation
#  to define the correlation structure
Id <- factor(rep(1,length(depth)))

# AR(1)
cor.gamm1 <- gamm(depth ~ s(distance, k=40, bs="ps", m=2),
                  correlation=corARMA(form=~distance|Id,p=1,q=0),
                  method="REML")
# AR(2)
cor.gamm2 <- gamm(depth ~ s(distance, k=40, bs="ps", m=2),
                  correlation=corARMA(form=~distance|Id,p=2,q=0),
                  method="REML")
```

The next Figure shows how including a correlation structure results in a smoother trend. In particular, the AR(2) model autocorrelation shows uncorrelated residuals.

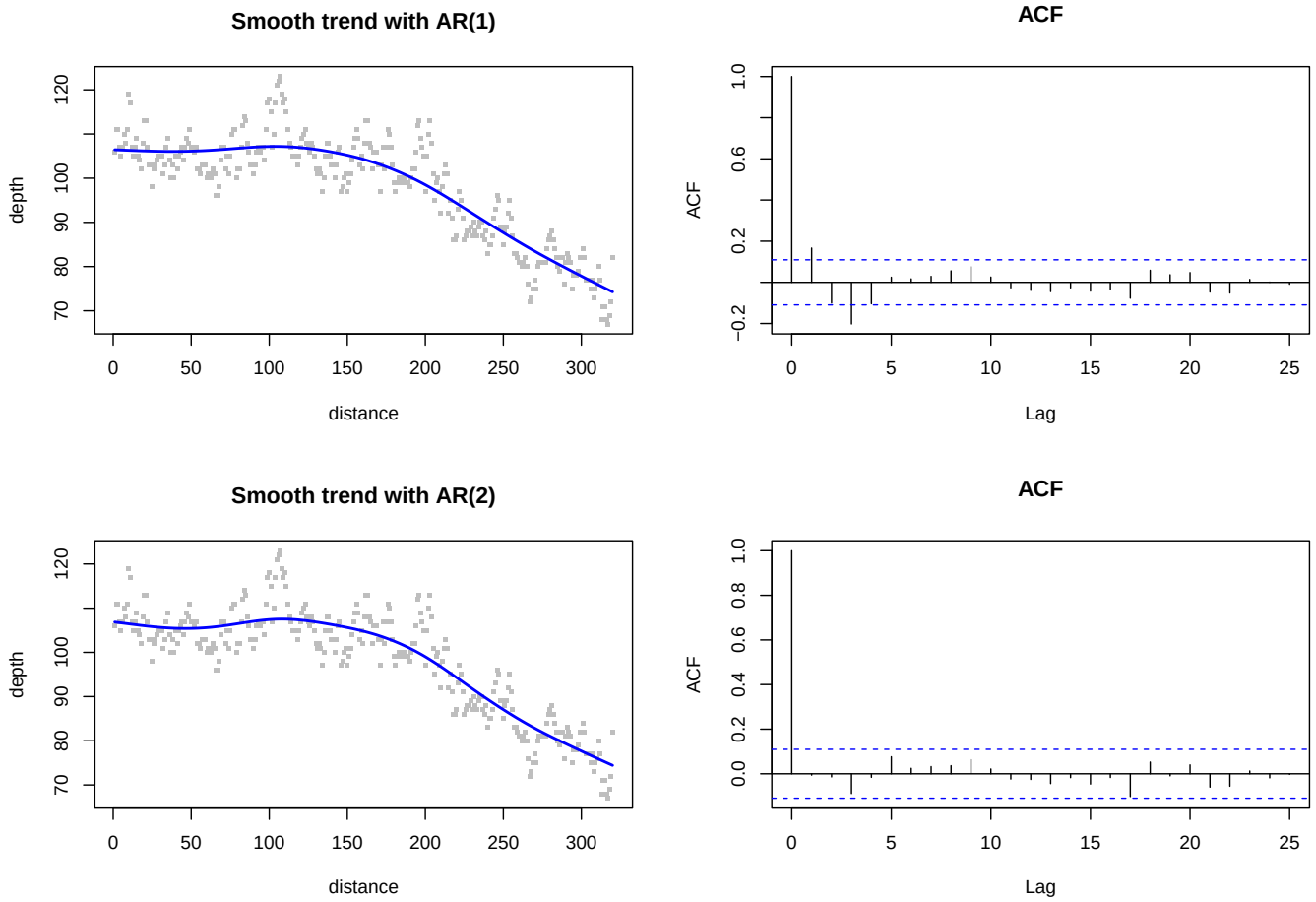



Figure 24: Smooth trends and ACF for AR(1) and AR(2) error models

## Check the smoothing parameters

```
print(c(cor.gamm0$gam$sp,cor.gamm1$gam$sp,cor.gamm2$gam$sp))
```

```
## s(distance) s(distance) s(distance)
##    2.117994 3228.963757 2014.146325
```

We can compare the model using anova function

```
anova(cor.gamm0$lme,cor.gamm1$lme,cor.gamm2$lme)
```

```
##                Model df      AIC      BIC    logLik   Test   L.Ratio p-value
## cor.gamm0$lme      1  4 1850.568 1865.616 -921.2841
## cor.gamm1$lme      2  5 1653.212 1672.022 -821.6058 1 vs 2 199.35667  <.0001
## cor.gamm2$lme      3  6 1638.010 1660.582 -813.0049 2 vs 3  17.20174  <.0001
```